Request for the development of a code MCNP_SOURCE_APP

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There is a need to develop software to automate the generation of accurate source terms for the code MCNP.

<u>Goal 1:</u> Read the entire ICRP-107 (2008) radionuclide emission database, and construct an MCNP format source definition file or files.

The format of the first ICRP-107 ASCII database (less than 14 MByte in size) is as follows (four nuclides are shown):

```
Ac-223        2.10m       480
 2 8.38497E+00 6.06280E-05  X
 2 2.14756E-05 1.04461E-04  X
 2 1.00091E-03 4.27494E-04  X
 2 7.83209E-07 6.59000E-04  X
 2 3.57877E-02 2.82466E-03  X
 2 1.76193E-06 2.93700E-03  X
 2 3.20904E-04 3.59600E-03  X
 2 2.32345E-03 1.04149E-02  X
 2 2.73926E-05 1.07215E-02  X
 2 2.47953E-05 1.13895E-02  X
 2 3.99189E-03 1.18951E-02  X
 2 3.51076E-02 1.20348E-02  X
 2 8.26792E-04 1.33519E-02  X
 2 5.61606E-04 1.39037E-02  X
 2 6.16982E-06 1.40413E-02  X
 2 6.55148E-06 1.42112E-02  X
 2 2.68951E-03 1.43175E-02  X
 2 3.82241E-05 1.43265E-02  X
 2 7.77346E-04 1.44364E-02  X
 2 7.07093E-03 1.44681E-02  X
 2 7.96536E-06 1.47640E-02  X
 2 4.42258E-05 1.47719E-02  X
 2 1.32777E-04 1.48016E-02  X
 2 2.99463E-02 1.48321E-02  X
 2 1.32159E-06 1.48530E-02  X
 2 1.44414E-06 1.48901E-02  X
 2 1.25882E-04 1.49653E-02  X
 2 1.13972E-03 1.49705E-02  X
 2 2.64542E-03 1.49855E-02  X
 1 2.47500E-05 1.50000E-02  G
 2 2.36379E-05 1.50014E-02  X
 2 1.90371E-07 1.50147E-02  X
 2 1.89135E-07 1.50199E-02  X
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2 8.25358E-07 1.50302E-02  X
2 1.10246E-04 1.54911E-02  X
2 1.64689E-04 1.56308E-02  X
2 2.19071E-04 1.68407E-02  X
2 1.10798E-05 1.71482E-02  X
2 6.45433E-03 1.73734E-02  X
2 7.10693E-04 1.76373E-02  X
2 3.37224E-05 1.77011E-02  X
2 5.22878E-05 1.77386E-02  X
2 7.79577E-04 1.78072E-02  X
2 2.45529E-06 1.78271E-02  X
2 1.06537E-03 1.79022E-02  X
2 9.32589E-06 1.79383E-02  X
2 3.27097E-07 1.79569E-02  X
2 3.25756E-07 1.79672E-02  X
2 1.78409E-05 1.80324E-02  X
2 2.84493E-05 1.80641E-02  X
2 1.57661E-04 1.84490E-02  X
2 1.73266E-04 1.84861E-02  X
2 2.75693E-06 1.85613E-02  X
2 4.41134E-06 1.85665E-02  X
2 2.37842E-05 1.86107E-02  X
2 2.30954E-05 1.86159E-02  X
1 2.77200E-04 3.57000E-02  G
1 2.37600E-04 4.11500E-02  G
1 1.28700E-04 4.24000E-02  G
1 3.76200E-04 5.69500E-02  G
1 1.98000E-04 6.44000E-02  G
1 1.08900E-04 6.60000E-02  G
1 9.90000E-05 7.28000E-02  G
1 9.90000E-05 7.28000E-02  G
1 5.44500E-04 7.82500E-02  G
1 9.90000E-05 8.24000E-02  G
2 7.43697E-03 8.35492E-02  X
1 5.74200E-03 8.35500E-02  G
2 1.22048E-02 8.64861E-02  X
1 4.95000E-04 8.96000E-02  G
1 3.86100E-03 9.27100E-02  G
2 1.38509E-03 9.72076E-02  X
2 2.69182E-03 9.78757E-02  X
2 4.29683E-05 9.83812E-02  X
2 5.05542E-05 9.85206E-02  X
1 8.91000E-03 9.85800E-02  G
2 3.38919E-04 1.00527E-01  X
2 6.74637E-04 1.00697E-01  X
2 1.20553E-05 1.00923E-01  X
2 1.41927E-05 1.00954E-01  X
1 3.96000E-04 1.01100E-01  G
2 8.05099E-05 1.01339E-01  X
2 1.54643E-04 1.01376E-01  X
2 2.09370E-06 1.01451E-01  X
2 2.43598E-06 1.01457E-01  X
2 1.12594E-05 1.01501E-01  X
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2 1.92773E-05 1.01506E-01  X
1 1.08900E-03 1.19400E-01  G
1 8.91000E-04 1.24800E-01  G
1 9.90000E-05 1.26400E-01  G
1 1.98000E-04 1.30500E-01  G
1 9.90000E-05 1.34600E-01  G
1 1.98000E-04 1.71700E-01  G
1 1.08900E-03 1.76300E-01  G
1 5.84100E-03 1.91300E-01  G
1 1.18800E-03 1.95400E-01  G
1 2.97000E-04 1.99300E-01  G
1 3.96000E-04 2.05700E-01  G
1 1.88100E-03 2.07000E-01  G
1 3.66300E-03 2.16000E-01  G
1 3.96000E-04 2.16100E-01  G
1 6.93000E-04 2.29700E-01  G
1 6.93000E-04 2.38100E-01  G
1 1.78200E-03 2.41700E-01  G
1 6.93000E-04 2.54400E-01  G
1 9.90000E-04 2.69200E-01  G
1 4.95000E-04 2.74000E-01  G
1 2.97000E-04 2.79800E-01  G
1 1.98000E-03 2.84200E-01  G
1 1.28700E-03 2.85700E-01  G
1 2.97000E-04 2.99100E-01  G
1 1.78200E-03 3.05500E-01  G
1 9.90000E-05 3.15200E-01  G
1 4.95000E-04 3.25300E-01  G
1 1.08900E-03 3.33500E-01  G
1 4.95000E-04 3.41000E-01  G
1 1.78200E-03 3.57400E-01  G
1 1.58400E-03 3.72400E-01  G
1 5.94000E-04 3.74800E-01  G
1 5.24700E-03 4.34200E-01  G
1 3.96000E-04 4.62200E-01  G
1 2.67300E-03 4.75200E-01  G
1 2.97000E-04 5.06600E-01  G
1 8.91000E-04 5.18800E-01  G
1 2.97000E-04 5.30000E-01  G
7 2.29727E+00 3.08263E-05 AE
7 2.36845E+00 1.19222E-04 AE
7 9.14825E-03 1.30284E-04 AE
7 1.36395E+00 1.83821E-04 AE
7 4.82933E-01 4.76605E-04 AE
7 5.54553E-02 1.30001E-03 AE
7 6.63206E-01 2.21694E-03 AE
7 1.07652E-01 2.60286E-03 AE
7 6.12931E-03 3.13154E-03 AE
7 7.74507E-02 9.25168E-03 AE
6 4.55461E-01 1.13555E-02 IE
7 3.70655E-02 1.20931E-02 AE
7 4.20982E-03 1.49724E-02 AE
6 1.50138E-01 1.50000E-02 IE
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6 1.16397E-02 1.70700E-02 IE
6 1.19345E-03 1.77290E-02 IE
6 8.12975E-03 1.78800E-02 IE
6 8.77733E-05 2.06660E-02 IE
6 4.07736E-03 2.25200E-02 IE
6 2.84049E-02 2.31790E-02 IE
6 5.86138E-03 2.32800E-02 IE
6 7.46374E-04 2.37700E-02 IE
6 2.63602E-02 2.44290E-02 IE
6 6.27952E-04 2.48800E-02 IE
6 2.70154E-02 2.61160E-02 IE
6 2.51667E-02 2.73660E-02 IE
6 3.09874E-03 3.20555E-02 IE
6 5.25188E-04 3.30800E-02 IE
6 1.05999E-03 3.57000E-02 IE
6 1.59088E-02 3.75055E-02 IE
6 5.37590E-05 3.83200E-02 IE
6 1.40772E-02 3.87555E-02 IE
6 4.11955E-05 3.89790E-02 IE
6 5.27967E-03 4.11500E-02 IE
6 4.43963E-05 4.19160E-02 IE
6 4.66419E-03 4.24000E-02 IE
6 1.65432E-04 4.57700E-02 IE
6 5.59629E-03 4.64290E-02 IE
6 8.20058E-05 4.73700E-02 IE
6 2.74305E-03 4.80290E-02 IE
6 4.80164E-03 4.93660E-02 IE
6 2.33725E-03 5.09660E-02 IE
6 3.39944E-05 5.33055E-02 IE
6 5.11745E-04 5.41700E-02 IE
6 5.00889E-05 5.41700E-02 IE
6 5.25358E-05 5.48290E-02 IE
6 1.57642E-03 5.48290E-02 IE
6 1.11180E-05 5.69500E-02 IE
6 1.30426E-03 5.77660E-02 IE
6 3.60191E-06 5.77660E-02 IE
6 2.27881E-03 5.96200E-02 IE
6 2.33993E-04 6.02790E-02 IE
6 2.85843E-03 6.07555E-02 IE
6 1.39748E-03 6.23555E-02 IE
6 1.59357E-05 6.32160E-02 IE
6 6.39618E-06 6.37700E-02 IE
6 9.50215E-04 6.44000E-02 IE
6 3.65922E-06 6.44290E-02 IE
6 1.98451E-02 6.49200E-02 IE
6 2.03945E-03 6.55790E-02 IE
6 4.64566E-04 6.60000E-02 IE
7 5.67808E-04 6.67586E-02 AE
6 3.59749E-06 6.73660E-02 IE
6 1.37951E-04 6.85160E-02 IE
6 1.35639E-04 6.91555E-02 IE
6 7.94262E-04 6.91555E-02 IE
6 2.01582E-05 7.01800E-02 IE
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6 2.65987E-05 7.09700E-02 IE
6 1.42692E-05 7.16290E-02 IE
6 4.65369E-05 7.28000E-02 IE
6 2.64189E-04 7.28000E-02 IE
6 1.92379E-04 7.40800E-02 IE
6 1.37434E-05 7.45660E-02 IE
6 6.04287E-04 7.46055E-02 IE
6 1.00571E-04 7.47390E-02 IE
6 3.78158E-04 7.47800E-02 IE
6 9.60574E-05 7.76760E-02 IE
6 2.07418E-04 7.82500E-02 IE
6 3.30698E-06 7.87555E-02 IE
6 5.25925E-03 7.99055E-02 IE
6 1.90388E-02 7.99500E-02 IE
6 1.95935E-03 8.06090E-02 IE
7 3.17685E-04 8.07477E-02 AE
6 1.09038E-06 8.24000E-02 IE
6 1.30209E-04 8.35460E-02 IE
6 1.80907E-03 8.35500E-02 IE
6 1.32070E-05 8.59555E-02 IE
6 9.40150E-05 8.90655E-02 IE
6 4.36235E-06 8.96000E-02 IE
6 1.64447E-03 8.97800E-02 IE
6 3.10797E-05 9.27100E-02 IE
6 8.92115E-05 9.38800E-02 IE
7 4.29565E-05 9.47537E-02 AE
6 5.05561E-03 9.49355E-02 IE
6 1.73614E-03 9.85800E-02 IE
6 1.33684E-03 1.00770E-01 IE
6 1.38009E-04 1.01429E-01 IE
6 2.63274E-05 1.04180E-01 IE
6 8.96662E-06 1.04366E-01 IE
6 2.96606E-03 1.05480E-01 IE
6 9.63155E-04 1.06170E-01 IE
6 9.94730E-05 1.06829E-01 IE
6 1.03175E-04 1.07770E-01 IE
6 1.06566E-05 1.08429E-01 IE
6 6.42825E-06 1.09766E-01 IE
6 6.87595E-07 1.11366E-01 IE
6 2.17174E-04 1.14480E-01 IE
6 5.53681E-04 1.14580E-01 IE
6 3.55876E-04 1.15756E-01 IE
6 8.62351E-05 1.15970E-01 IE
6 8.90266E-06 1.16629E-01 IE
6 1.21888E-04 1.19400E-01 IE
6 5.70024E-07 1.19566E-01 IE
6 2.56309E-04 1.21156E-01 IE
6 2.74524E-05 1.22756E-01 IE
6 8.77942E-05 1.24800E-01 IE
6 9.40361E-06 1.26400E-01 IE
6 3.55984E-05 1.28180E-01 IE
6 2.29071E-05 1.30956E-01 IE
6 7.84905E-06 1.34600E-01 IE
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6 3.27534E-05 1.36580E-01 IE
6 8.13511E-05 1.40180E-01 IE
6 6.14047E-04 1.52880E-01 IE
6 2.47017E-06 1.53070E-01 IE
6 8.30404E-07 1.53729E-01 IE
6 6.79227E-07 1.56666E-01 IE
6 9.47970E-05 1.57670E-01 IE
6 1.38048E-05 1.58329E-01 IE
6 1.48980E-05 1.61266E-01 IE
6 7.49738E-04 1.67680E-01 IE
6 9.52912E-07 1.68056E-01 IE
6 3.18228E-07 1.71700E-01 IE
6 1.69550E-05 1.72480E-01 IE
6 3.18284E-05 1.72656E-01 IE
6 3.97836E-04 1.72670E-01 IE
6 5.89401E-05 1.73329E-01 IE
6 5.75742E-05 1.76266E-01 IE
6 1.10137E-05 1.76300E-01 IE
6 1.10502E-05 1.76770E-01 IE
6 3.40821E-06 1.77429E-01 IE
6 9.70159E-06 1.78280E-01 IE
6 2.69825E-06 1.80366E-01 IE
6 6.24378E-05 1.82680E-01 IE
6 4.01058E-05 1.84180E-01 IE
6 3.28180E-06 1.87070E-01 IE
6 1.32004E-04 1.87656E-01 IE
6 9.77676E-07 1.87729E-01 IE
6 4.84286E-04 1.88370E-01 IE
6 4.96224E-05 1.89029E-01 IE
6 7.64083E-07 1.90666E-01 IE
6 4.55724E-05 1.91300E-01 IE
6 4.10947E-06 1.91756E-01 IE
6 3.01609E-06 1.91966E-01 IE
6 1.37398E-06 1.95400E-01 IE
6 2.72100E-05 1.97370E-01 IE
6 9.03528E-05 1.97470E-01 IE
6 8.35116E-06 1.97580E-01 IE
6 7.84288E-06 1.98029E-01 IE
6 9.24688E-06 1.98129E-01 IE
6 6.05509E-06 2.00966E-01 IE
6 5.59273E-07 2.01066E-01 IE
6 1.20297E-06 2.02056E-01 IE
6 1.27944E-04 2.03356E-01 IE
6 9.54360E-04 2.03980E-01 IE
6 4.02422E-07 2.05700E-01 IE
6 4.39214E-05 2.07000E-01 IE
6 4.48827E-06 2.11070E-01 IE
6 1.24145E-06 2.11729E-01 IE
6 9.84153E-06 2.12356E-01 IE
6 2.38710E-05 2.12456E-01 IE
6 4.86900E-05 2.13680E-01 IE
6 9.43680E-07 2.14666E-01 IE
6 3.29393E-06 2.16000E-01 IE
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6 8.19375E-06 2.16100E-01 IE
6 4.14587E-06 2.19470E-01 IE
6 1.11906E-06 2.20129E-01 IE
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6 1.03132E-05 2.23070E-01 IE
6 2.75569E-06 2.23729E-01 IE
6 1.15530E-05 2.23780E-01 IE
6 1.59711E-06 2.26056E-01 IE
6 2.06808E-06 2.26666E-01 IE
6 5.34925E-07 2.29700E-01 IE
6 4.59423E-04 2.31980E-01 IE
6 1.46088E-06 2.34456E-01 IE
6 1.00127E-04 2.35770E-01 IE
6 1.02148E-05 2.36429E-01 IE
6 3.61978E-06 2.38056E-01 IE
6 4.89660E-07 2.38100E-01 IE
6 6.06946E-07 2.39366E-01 IE
6 1.21358E-06 2.41700E-01 IE
6 1.22190E-04 2.50570E-01 IE
6 2.64913E-05 2.50756E-01 IE
6 1.24473E-05 2.51229E-01 IE
6 7.35352E-07 2.54166E-01 IE
6 9.07458E-06 2.54400E-01 IE
6 2.17639E-06 2.55370E-01 IE
6 3.38751E-05 2.55880E-01 IE
6 5.34148E-07 2.56029E-01 IE
6 3.88567E-07 2.58966E-01 IE
6 1.24779E-06 2.61170E-01 IE
6 3.01905E-07 2.61829E-01 IE
6 2.18497E-07 2.64766E-01 IE
6 3.23461E-05 2.65556E-01 IE
6 8.04231E-06 2.65570E-01 IE
6 1.92525E-06 2.66229E-01 IE
6 5.16836E-06 2.67070E-01 IE
6 1.23281E-06 2.67729E-01 IE
6 1.38805E-06 2.69166E-01 IE
6 1.10710E-05 2.69200E-01 IE
6 7.39880E-07 2.70356E-01 IE
6 8.87684E-07 2.70666E-01 IE
6 2.75644E-05 2.70880E-01 IE
6 1.82695E-04 2.73280E-01 IE
6 2.48581E-07 2.74000E-01 IE
6 4.22010E-07 2.76156E-01 IE
6 1.41835E-07 2.79800E-01 IE
6 1.08060E-06 2.80470E-01 IE
6 2.70958E-06 2.80556E-01 IE
6 2.49790E-07 2.81129E-01 IE
6 1.73907E-06 2.82056E-01 IE
6 1.77875E-07 2.84066E-01 IE
6 9.10919E-07 2.84200E-01 IE
6 5.84700E-07 2.85700E-01 IE
6 1.54906E-04 2.86870E-01 IE
6 1.57079E-05 2.87529E-01 IE
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6 9.16946E-07 2.90466E-01 IE
6 3.59603E-07 2.95456E-01 IE
6 7.89524E-06 2.96570E-01 IE
6 7.99425E-07 2.97229E-01 IE
6 1.20998E-07 2.99100E-01 IE
6 4.65384E-08 3.00166E-01 IE
6 4.10384E-05 3.01856E-01 IE
6 1.40233E-05 3.05500E-01 IE
6 1.50623E-06 3.06670E-01 IE
6 3.28489E-07 3.07329E-01 IE
6 2.29312E-07 3.10266E-01 IE
6 2.09173E-06 3.11556E-01 IE
6 7.43956E-05 3.14870E-01 IE
6 7.14555E-07 3.15200E-01 IE
6 7.50685E-06 3.15529E-01 IE
6 4.34815E-07 3.18466E-01 IE
6 4.91384E-07 3.21656E-01 IE
6 1.65566E-07 3.25300E-01 IE
6 1.96973E-05 3.29856E-01 IE
6 1.08410E-03 3.32680E-01 IE
6 6.72913E-06 3.33500E-01 IE
6 4.45605E-06 3.38770E-01 IE
6 9.08809E-07 3.39429E-01 IE
6 6.21073E-07 3.42366E-01 IE
6 1.42213E-06 3.53756E-01 IE
6 3.63906E-06 3.53770E-01 IE
6 7.20618E-07 3.54429E-01 IE
6 2.95248E-05 3.56170E-01 IE
6 2.95219E-06 3.56829E-01 IE
6 4.87996E-07 3.57366E-01 IE
6 4.79815E-07 3.57400E-01 IE
6 1.69400E-07 3.59766E-01 IE
6 4.39210E-06 3.60680E-01 IE
6 1.15066E-06 3.68756E-01 IE
6 7.79819E-06 3.71156E-01 IE
6 3.88437E-07 3.72400E-01 IE
6 4.33744E-04 3.73680E-01 IE
6 2.66647E-06 3.74800E-01 IE
6 2.73505E-06 4.05080E-01 IE
6 1.75247E-04 4.15570E-01 IE
6 1.72705E-05 4.16229E-01 IE
6 1.14503E-04 4.17280E-01 IE
6 9.82667E-07 4.19166E-01 IE
6 2.49775E-06 4.28480E-01 IE
6 4.61082E-05 4.30556E-01 IE
6 1.57897E-05 4.34200E-01 IE
6 5.88410E-07 4.43570E-01 IE
6 9.96848E-08 4.44229E-01 IE
6 6.44371E-08 4.47166E-01 IE
6 7.00575E-05 4.56570E-01 IE
6 6.83177E-06 4.57229E-01 IE
6 1.77878E-07 4.58556E-01 IE
6 3.87723E-07 4.60166E-01 IE
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6 6.01905E-08 4.62200E-01 IE
6 1.83897E-05 4.71556E-01 IE
6 6.30119E-06 4.75200E-01 IE
6 3.68310E-07 4.87970E-01 IE
6 5.84659E-08 4.88629E-01 IE
6 3.70196E-08 4.91566E-01 IE
6 1.84610E-05 5.00170E-01 IE
6 1.77983E-06 5.00829E-01 IE
6 1.09597E-07 5.02955E-01 IE
6 1.01003E-07 5.03766E-01 IE
6 3.71023E-08 5.06600E-01 IE
6 3.37296E-07 5.11370E-01 IE
6 5.18511E-08 5.12029E-01 IE
6 3.25073E-08 5.14966E-01 IE
6 4.83684E-06 5.15155E-01 IE
6 1.65722E-06 5.18800E-01 IE
6 9.96333E-08 5.26356E-01 IE
6 3.37352E-08 5.30000E-01 IE
8 9.77913E-05 5.89732E+00  A
8 2.93374E-04 5.96852E+00  A
8 9.77913E-05 6.02302E+00  A
8 2.93374E-04 6.08292E+00  A
8 1.17350E-03 6.13713E+00  A
8 2.93374E-04 6.14087E+00  A
8 4.88956E-04 6.16394E+00  A
8 9.29017E-03 6.17985E+00  A
8 2.93374E-04 6.20745E+00  A
8 5.86748E-05 6.22434E+00  A
8 8.80122E-04 6.23711E+00  A
8 4.88956E-04 6.28395E+00  A
8 5.86748E-04 6.29328E+00  A
8 4.59619E-03 6.29564E+00  A
8 2.93374E-03 6.32716E+00  A
8 1.36908E-03 6.33384E+00  A
8 4.88956E-04 6.34219E+00  A
8 2.15141E-03 6.36134E+00  A
8 1.27129E-03 6.39699E+00  A
8 1.95583E-03 6.44923E+00  A
8 5.86748E-04 6.45473E+00  A
8 3.03153E-02 6.47350E+00  A
8 5.86748E-03 6.52406E+00  A
8 3.03153E-02 6.52937E+00  A
8 1.33974E-01 6.56454E+00  A
8 2.93374E-03 6.58181E+00  A
8 9.77913E-03 6.60626E+00  A
8 4.36149E-01 6.64662E+00  A
8 3.10976E-01 6.66135E+00  A
9 9.77913E-05 1.07784E-01 AR
9 2.93374E-04 1.09085E-01 AR
9 9.77913E-05 1.10081E-01 AR
9 2.93374E-04 1.11176E-01 AR
9 1.17350E-03 1.12167E-01 AR
9 2.93374E-04 1.12235E-01 AR
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 9 4.88956E-04 1.12657E-01 AR
 9 9.29017E-03 1.12947E-01 AR
 9 2.93374E-04 1.13452E-01 AR
 9 5.86748E-05 1.13760E-01 AR
 9 8.80122E-04 1.13994E-01 AR
 9 4.88956E-04 1.14850E-01 AR
 9 5.86748E-04 1.15020E-01 AR
 9 4.59619E-03 1.15064E-01 AR
 9 2.93374E-03 1.15640E-01 AR
 9 1.36908E-03 1.15762E-01 AR
 9 4.88956E-04 1.15914E-01 AR
 9 2.15141E-03 1.16264E-01 AR
 9 1.27129E-03 1.16916E-01 AR
 9 1.95583E-03 1.17871E-01 AR
 9 5.86748E-04 1.17971E-01 AR
 9 3.03153E-02 1.18314E-01 AR
 9 5.86748E-03 1.19238E-01 AR
 9 3.03153E-02 1.19335E-01 AR
 9 1.33974E-01 1.19978E-01 AR
 9 2.93374E-03 1.20294E-01 AR
 9 9.77913E-03 1.20741E-01 AR
 9 4.36149E-01 1.21478E-01 AR
 9 3.10976E-01 1.21748E-01 AR
Ac-224        2.78h       682
 2 1.26364E+00 6.05941E-05  X
 2 1.52234E+01 7.12229E-05  X
 2 3.23790E-06 1.04068E-04  X
 2 4.64062E-05 1.11439E-04  X
 2 1.43848E-04 4.30159E-04  X
 2 1.76238E-03 4.31588E-04  X
 2 4.53773E-07 6.59000E-04  X
 2 2.11568E-06 6.72000E-04  X
 2 5.58834E-03 2.81326E-03  X
 2 7.80547E-02 2.88379E-03  X
 2 1.99037E-07 2.93700E-03  X
 2 9.40375E-06 3.10400E-03  X
 2 1.85924E-04 3.59600E-03  X
 2 9.53478E-04 3.77600E-03  X
 1 1.67277E-08 6.90000E-03  G
 2 5.64293E-04 1.04149E-02  X
 2 1.26864E-02 1.06609E-02  X
 2 6.65281E-06 1.07215E-02  X
 2 1.51937E-04 1.09740E-02  X
 2 6.02199E-06 1.13895E-02  X
 2 1.37643E-04 1.16823E-02  X
 2 9.69505E-04 1.18951E-02  X
 2 8.52652E-03 1.20348E-02  X
 2 2.15923E-02 1.21973E-02  X
 2 1.89877E-01 1.23447E-02  X
 1 4.35712E-05 1.24000E-02  G
 1 1.05090E-05 1.30000E-02  G
 2 9.33990E-05 1.33519E-02  X
 2 3.97170E-03 1.37649E-02  X
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2 1.36396E-04 1.39037E-02  X
2 1.49846E-06 1.40413E-02  X
2 1.59115E-06 1.42112E-02  X
2 3.09757E-03 1.42663E-02  X
2 1.55824E-03 1.43175E-02  X
2 4.31801E-06 1.43265E-02  X
2 3.41748E-05 1.44074E-02  X
2 1.88793E-04 1.44364E-02  X
2 1.71731E-03 1.44681E-02  X
2 3.65913E-05 1.45887E-02  X
2 7.56662E-03 1.47500E-02  X
2 1.93454E-06 1.47640E-02  X
2 1.07411E-05 1.47719E-02  X
2 1.87776E-04 1.47863E-02  X
2 3.22473E-05 1.48016E-02  X
2 4.23501E-03 1.48192E-02  X
2 3.38290E-03 1.48321E-02  X
2 3.85683E-02 1.48529E-02  X
2 3.20972E-07 1.48530E-02  X
2 3.50736E-07 1.48900E-02  X
2 3.05727E-05 1.49653E-02  X
2 2.76802E-04 1.49705E-02  X
2 1.53269E-03 1.49855E-02  X
2 5.74090E-06 1.50013E-02  X
2 4.62351E-08 1.50147E-02  X
2 4.59349E-08 1.50199E-02  X
2 2.00453E-07 1.50302E-02  X
2 4.54885E-05 1.51563E-02  X
2 2.52215E-04 1.51648E-02  X
2 7.44848E-04 1.52024E-02  X
2 7.46716E-06 1.52557E-02  X
2 8.15084E-06 1.52962E-02  X
2 1.43379E-01 1.53013E-02  X
2 7.16367E-04 1.53747E-02  X
2 6.50617E-03 1.53805E-02  X
2 1.41565E-04 1.54164E-02  X
2 1.18747E-06 1.54311E-02  X
2 1.21902E-06 1.54374E-02  X
2 9.68762E-06 1.54512E-02  X
2 7.26896E-03 1.54583E-02  X
2 6.38737E-05 1.54911E-02  X
2 9.54170E-05 1.56308E-02  X
2 3.17709E-04 1.59733E-02  X
1 3.75224E-05 1.60000E-02  G
2 4.74419E-04 1.61207E-02  X
2 2.47475E-05 1.68407E-02  X
2 1.25164E-06 1.71482E-02  X
2 1.05775E-03 1.73703E-02  X
2 7.29117E-04 1.73734E-02  X
2 4.11759E-04 1.76373E-02  X
2 5.48866E-05 1.76927E-02  X
2 3.80947E-06 1.77010E-02  X
2 5.90672E-06 1.77386E-02  X
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2 4.51668E-04 1.78072E-02  X
2 2.77363E-07 1.78271E-02  X
2 1.20350E-04 1.79022E-02  X
2 3.12240E-02 1.79232E-02  X
2 1.05350E-06 1.79384E-02  X
2 3.69507E-08 1.79569E-02  X
2 3.67992E-08 1.79672E-02  X
2 1.03366E-05 1.80324E-02  X
2 1.64829E-05 1.80641E-02  X
2 2.00826E-03 1.81834E-02  X
2 1.69429E-04 1.82603E-02  X
2 2.56838E-04 1.83063E-02  X
2 2.16346E-03 1.83647E-02  X
2 1.24701E-05 1.84001E-02  X
2 9.13448E-05 1.84490E-02  X
2 5.38870E-03 1.84787E-02  X
2 1.00386E-04 1.84861E-02  X
2 4.89138E-05 1.85205E-02  X
2 1.87328E-06 1.85414E-02  X
2 2.43158E-06 1.85552E-02  X
2 1.59730E-06 1.85613E-02  X
2 2.55583E-06 1.85665E-02  X
2 5.19461E-05 1.85952E-02  X
2 1.37800E-05 1.86107E-02  X
2 1.33809E-05 1.86159E-02  X
2 8.28975E-05 1.86289E-02  X
2 4.55098E-04 1.90317E-02  X
2 4.93044E-04 1.90722E-02  X
2 8.37920E-06 1.91507E-02  X
2 1.34629E-05 1.91565E-02  X
2 7.77151E-05 1.92071E-02  X
2 7.55197E-05 1.92134E-02  X
1 5.87165E-06 2.04000E-02  G
1 8.64081E-05 2.46000E-02  G
1 1.82000E-05 3.59000E-02  G
1 2.73000E-05 3.59000E-02  G
1 5.46000E-05 3.73000E-02  G
1 4.55000E-05 4.04000E-02  G
1 3.64000E-05 4.15000E-02  G
1 2.73000E-05 4.45000E-02  G
1 2.73000E-05 4.60000E-02  G
1 5.91500E-04 4.84000E-02  G
1 3.64000E-05 4.98000E-02  G
1 9.10000E-05 5.31000E-02  G
1 6.37000E-05 5.43000E-02  G
1 3.64000E-05 5.53000E-02  G
1 4.18600E-04 6.06700E-02  G
1 2.09300E-04 6.60000E-02  G
1 5.46000E-04 6.66000E-02  G
1 7.28000E-05 7.09000E-02  G
1 4.55000E-05 7.29000E-02  G
1 1.27400E-04 7.29000E-02  G
1 5.46000E-05 7.36000E-02  G
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1 5.99940E-05 7.44000E-02  G
1 2.54800E-04 7.91000E-02  G
1 1.91100E-04 8.15000E-02  G
2 1.59761E-03 8.35492E-02  X
1 1.63800E-03 8.39000E-02  G
1 1.51985E-02 8.44000E-02  G
2 2.25200E-01 8.57602E-02  X
2 2.62183E-03 8.64861E-02  X
1 1.18300E-04 8.74000E-02  G
2 3.67781E-01 8.88641E-02  X
1 4.73200E-04 9.23400E-02  G
2 2.97546E-04 9.72076E-02  X
2 5.78256E-04 9.78757E-02  X
2 9.23044E-06 9.83812E-02  X
2 1.08601E-05 9.85206E-02  X
2 4.18857E-02 9.98380E-02  X
2 7.28066E-05 1.00527E-01  X
2 8.15171E-02 1.00546E-01  X
2 1.44925E-04 1.00697E-01  X
2 2.58971E-06 1.00923E-01  X
2 3.04888E-06 1.00954E-01  X
2 1.33176E-03 1.01061E-01  X
2 1.55716E-03 1.01209E-01  X
2 1.72951E-05 1.01339E-01  X
2 3.32203E-05 1.01376E-01  X
2 4.49769E-07 1.01451E-01  X
2 5.23297E-07 1.01457E-01  X
2 2.41875E-06 1.01501E-01  X
2 4.14115E-06 1.01506E-01  X
1 3.64000E-05 1.03100E-01  G
2 1.03002E-02 1.03272E-01  X
2 2.05603E-02 1.03453E-01  X
2 3.77516E-04 1.03683E-01  X
2 4.41986E-04 1.03717E-01  X
2 2.46000E-03 1.04120E-01  X
2 4.75607E-03 1.04160E-01  X
2 6.85807E-05 1.04239E-01  X
2 7.95575E-05 1.04245E-01  X
2 3.86259E-04 1.04295E-01  X
2 6.77960E-04 1.04301E-01  X
1 1.18300E-04 1.04400E-01  G
1 9.55500E-04 1.08430E-01  G
1 2.73000E-05 1.14700E-01  G
1 2.73000E-05 1.16500E-01  G
1 3.64000E-05 1.22300E-01  G
1 3.64000E-05 1.23300E-01  G
1 4.55000E-05 1.26700E-01  G
1 7.37100E-04 1.28400E-01  G
1 3.27600E-04 1.28900E-01  G
1 2.68973E-01 1.31350E-01  G
1 3.23960E-03 1.40700E-01  G
1 9.64600E-04 1.41400E-01  G
1 2.04750E-03 1.44440E-01  G
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1 1.13750E-03 1.49900E-01  G
1 7.37100E-03 1.56820E-01  G
1 1.45600E-04 1.58500E-01  G
1 4.55000E-05 1.63300E-01  G
1 4.55000E-04 1.64800E-01  G
1 2.69973E-03 1.66400E-01  G
1 3.64000E-05 1.70200E-01  G
1 5.46000E-05 1.86800E-01  G
1 6.37000E-05 1.94500E-01  G
1 2.73000E-04 1.99500E-01  G
1 5.46000E-05 2.01100E-01  G
1 3.19968E-03 2.05900E-01  G
1 3.27600E-04 2.07600E-01  G
1 1.82000E-05 2.10000E-01  G
1 1.82000E-05 2.11500E-01  G
1 5.22948E-01 2.15750E-01  G
1 2.91200E-04 2.25300E-01  G
1 5.46000E-05 2.27700E-01  G
1 2.36600E-04 2.51600E-01  G
1 2.73000E-05 2.58800E-01  G
1 1.72900E-03 2.61300E-01  G
1 1.18300E-04 2.63100E-01  G
1 9.10000E-05 2.71500E-01  G
1 1.00100E-04 2.73700E-01  G
1 6.37000E-05 2.87600E-01  G
1 2.73000E-05 2.91200E-01  G
1 1.54700E-04 3.00000E-01  G
1 6.37000E-05 3.28000E-01  G
1 6.37000E-05 3.28000E-01  G
1 4.55000E-05 3.40300E-01  G
1 2.73000E-05 3.57000E-01  G
1 1.91100E-04 3.64100E-01  G
1 2.73000E-05 3.76500E-01  G
1 4.55000E-05 3.82300E-01  G
1 2.73000E-05 4.09800E-01  G
1 2.73000E-05 4.15800E-01  G
1 2.73000E-05 4.24400E-01  G
1 3.64000E-05 4.91200E-01  G
7 4.18710E+00 2.56146E-05 AE
7 3.47004E-01 3.06891E-05 AE
7 4.23277E+00 1.20249E-04 AE
7 3.57638E-01 1.20647E-04 AE
7 1.38100E-03 1.30860E-04 AE
7 1.85380E-02 1.35371E-04 AE
7 2.39001E+00 1.81185E-04 AE
7 1.99628E-01 1.84984E-04 AE
7 6.03470E-01 4.63898E-04 AE
7 7.10897E-02 5.05362E-04 AE
6 1.38165E-04 9.66000E-04 IE
7 2.79168E-02 1.15505E-03 AE
7 1.67864E-01 1.43391E-03 AE
6 1.24885E-05 1.58000E-03 IE
6 1.29841E-03 1.77000E-03 IE
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7 1.01816E-01 2.19015E-03 AE
7 1.35075E+00 2.21570E-03 AE
6 1.33608E-04 2.42900E-03 IE
7 1.61157E-02 2.56102E-03 AE
7 2.16950E-01 2.58614E-03 AE
6 3.94174E-05 2.88001E-03 IE
7 9.02017E-04 3.07991E-03 AE
7 1.21454E-02 3.11171E-03 AE
6 1.47327E-02 3.25550E-03 IE
6 1.02778E-05 5.36600E-03 IE
6 1.09059E-02 5.97000E-03 IE
6 1.12270E-03 6.62900E-03 IE
6 4.83232E-03 6.90000E-03 IE
6 2.90645E-04 6.91000E-03 IE
6 1.12941E-02 8.75550E-03 IE
7 1.79568E-02 9.21579E-03 AE
7 3.68176E-01 9.31739E-03 AE
6 2.36584E-03 9.35550E-03 IE
6 8.52880E-05 9.56600E-03 IE
7 8.58944E-03 1.20650E-02 AE
7 1.78326E-01 1.22421E-02 AE
6 4.54694E-03 1.23555E-02 IE
6 3.85932E-03 1.24000E-02 IE
6 8.08654E-04 1.30000E-02 IE
6 7.27388E-06 1.31800E-02 IE
7 9.75940E-04 1.49600E-02 AE
7 2.04479E-02 1.51976E-02 AE
6 1.55637E-03 1.60000E-02 IE
6 3.44609E-04 1.67555E-02 IE
6 7.51670E-04 1.72700E-02 IE
6 1.12751E-03 1.72700E-02 IE
6 1.15613E-04 1.79290E-02 IE
6 7.70753E-05 1.79290E-02 IE
6 2.01335E-03 1.86700E-02 IE
6 2.06511E-04 1.93290E-02 IE
6 1.18221E-04 2.04000E-02 IE
6 5.66557E-06 2.08660E-02 IE
6 8.49836E-06 2.08660E-02 IE
6 2.90958E-03 2.09555E-02 IE
6 1.29338E-05 2.17700E-02 IE
6 1.51253E-05 2.22660E-02 IE
6 1.33882E-05 2.24290E-02 IE
6 9.80386E-04 2.28700E-02 IE
6 1.00390E-04 2.35290E-02 IE
6 9.97110E-04 2.46000E-02 IE
6 1.57614E-05 2.53660E-02 IE
6 7.27352E-06 2.64660E-02 IE
6 1.50476E-04 2.68800E-02 IE
6 5.27477E-02 2.70300E-02 IE
6 6.62975E-05 2.73800E-02 IE
6 1.01202E-02 2.97700E-02 IE
6 1.03275E-03 3.04290E-02 IE
6 6.83719E-06 3.11700E-02 IE
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6 1.18226E-03 3.44700E-02 IE
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6 1.20849E-04 3.51290E-02 IE
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6 1.02683E-04 3.59000E-02 IE
6 6.84555E-05 3.59000E-02 IE
6 7.92130E-05 3.63290E-02 IE
6 5.52842E-06 3.66700E-02 IE
6 1.03968E-05 3.67555E-02 IE
6 1.83411E-04 3.73000E-02 IE
6 4.33839E-06 3.73290E-02 IE
6 2.60308E-04 3.78555E-02 IE
6 8.59392E-06 3.80660E-02 IE
6 5.30870E-04 3.91800E-02 IE
6 5.62186E-06 3.92660E-02 IE
6 1.56145E-04 3.98800E-02 IE
6 4.71394E-06 4.02660E-02 IE
6 3.37060E-06 4.04000E-02 IE
6 8.92315E-05 4.15000E-02 IE
6 3.67868E-03 4.20400E-02 IE
6 3.77491E-04 4.26990E-02 IE
6 3.15188E-04 4.29200E-02 IE
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6 2.64082E-05 4.56360E-02 IE
6 4.72631E-06 4.61555E-02 IE
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6 1.54092E-06 4.98000E-02 IE
6 2.05997E-04 5.06555E-02 IE
6 1.02380E-05 5.09660E-02 IE
6 3.92055E-05 5.15660E-02 IE
6 3.56106E-06 5.16555E-02 IE
6 1.07886E-04 5.31000E-02 IE
6 2.28974E-05 5.42700E-02 IE
6 6.55909E-04 5.42700E-02 IE
6 7.07137E-05 5.43000E-02 IE
6 7.19894E-04 5.49290E-02 IE
6 6.73376E-05 5.49290E-02 IE
6 2.73355E-04 5.49700E-02 IE
6 3.16118E-05 5.51680E-02 IE
6 1.16388E-06 5.53000E-02 IE
6 9.30079E-04 5.53000E-02 IE
6 2.80689E-05 5.56290E-02 IE
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6 9.43589E-04 5.58400E-02 IE
6 1.78948E-05 5.69800E-02 IE
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6 5.95353E-04 5.78660E-02 IE
6 4.61603E-06 5.78660E-02 IE
6 1.92209E-06 5.85660E-02 IE
6 7.59462E-04 5.89440E-02 IE
6 5.21447E-04 6.04700E-02 IE
6 3.35430E-04 6.06700E-02 IE
6 5.35515E-05 6.11290E-02 IE
6 5.22490E-06 6.17800E-02 IE
6 6.07001E-04 6.20800E-02 IE
6 3.81950E-04 6.23555E-02 IE
6 1.26471E-05 6.28700E-02 IE
6 3.23405E-05 6.29555E-02 IE
6 5.11092E-05 6.32800E-02 IE
6 7.29712E-06 6.35290E-02 IE
6 3.64296E-06 6.40660E-02 IE
6 4.98572E-03 6.51680E-02 IE
6 1.01724E-04 6.52700E-02 IE
6 1.33305E-01 6.58400E-02 IE
6 5.73898E-05 6.59290E-02 IE
6 1.31003E-04 6.60000E-02 IE
6 7.19337E-06 6.64660E-02 IE
6 1.06131E-05 6.66000E-02 IE
7 1.21977E-04 6.67586E-02 AE
7 1.68070E-02 6.84262E-02 AE
6 3.07389E-05 6.87700E-02 IE
6 5.61728E-05 6.88660E-02 IE
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6 3.62657E-04 6.92555E-02 IE
6 1.73851E-04 6.92555E-02 IE
6 8.08787E-04 6.94290E-02 IE
6 7.24565E-05 6.99555E-02 IE
6 4.72153E-04 7.06263E-02 IE
6 6.28801E-04 7.23660E-02 IE
6 1.20627E-04 7.29000E-02 IE
6 5.96477E-05 7.29000E-02 IE
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6 2.37877E-05 7.37100E-02 IE
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6 4.74695E-05 7.91000E-02 IE
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6 6.57192E-02 8.06263E-02 IE
7 6.82452E-05 8.07477E-02 AE
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6 1.71916E-05 8.39000E-02 IE
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6 9.36484E-08 4.87556E-01 IE
6 3.15797E-08 4.91200E-01 IE
8 5.47973E-05 5.64431E+00  A
8 1.09595E-04 5.71188E+00  A
8 1.09595E-04 5.72160E+00  A
8 4.56644E-05 5.74282E+00  A
8 2.19189E-04 5.76953E+00  A
8 5.47973E-05 5.77955E+00  A
8 1.82658E-05 5.80666E+00  A
8 2.37455E-04 5.84054E+00  A
8 5.02308E-04 5.84407E+00  A
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8 2.28322E-04 5.85920E+00  A
8 6.84966E-04 5.86499E+00  A
8 9.13288E-05 5.87216E+00  A
8 1.55259E-03 5.87963E+00  A
8 1.27860E-04 5.90467E+00  A
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8 8.76756E-04 5.91921E+00  A
8 4.01847E-03 5.94504E+00  A
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8 1.64392E-04 5.97146E+00  A
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 9 1.08681E-02 1.12929E-01 AR
 9 1.86311E-02 1.13052E-01 AR
Sr-90        28.79y         1
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Y-90         64.10h        65
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 2 9.74822E-10 2.37008E-05  X
 2 1.30841E-13 8.73999E-05  X
 2 6.12675E-08 1.80157E-04  X
 2 3.18714E-10 2.02400E-04  X
 2 1.99114E-09 2.89800E-04  X
 2 3.80113E-07 1.80274E-03  X
 2 1.52314E-09 1.87987E-03  X
 2 1.10461E-07 1.89014E-03  X
 2 1.43958E-09 1.89392E-03  X
 2 9.81094E-10 1.98132E-03  X
 2 2.30294E-07 2.03171E-03  X
 2 2.09577E-06 2.03430E-03  X
 2 9.70357E-07 2.11911E-03  X
 2 1.74044E-08 2.16484E-03  X
 2 3.70386E-08 2.16967E-03  X
 2 6.44656E-08 2.18372E-03  X
 2 3.95812E-11 2.18617E-03  X
 2 3.79503E-11 2.18806E-03  X
 2 2.86922E-09 2.21813E-03  X
 2 2.54092E-08 2.21828E-03  X
 2 7.58913E-09 2.25224E-03  X
 2 3.29324E-11 2.27546E-03  X
 2 1.21910E-08 2.30552E-03  X
 2 2.84184E-10 2.32150E-03  X
 2 4.23154E-10 2.32411E-03  X
 2 5.35896E-09 2.47597E-03  X
 2 9.41235E-09 2.47787E-03  X
 2 3.13925E-12 2.50793E-03  X
 2 4.67531E-12 2.50808E-03  X
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 2 4.12457E-05 1.57372E-02  X
 2 3.25276E-06 1.76171E-02  X
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 2 1.02298E-06 1.79252E-02  X
 2 1.91513E-10 1.79553E-02  X
 2 2.70221E-10 1.79555E-02  X
 1 1.40000E-08 2.18624E+00  G
 5 1.40000E-08 2.46152E-02 B-
 5 1.15000E-04 1.85079E-01 B-
 5 9.99885E-01 9.32991E-01 B-
 7 7.35809E-04 2.22364E-05 AE
 7 1.09988E-04 9.84441E-05 AE
 7 2.27673E-05 1.06128E-04 AE
 7 2.52113E-04 1.17862E-04 AE
 7 7.44393E-06 1.58122E-04 AE
 7 1.04772E-04 1.75371E-03 AE
 7 1.17454E-05 1.94448E-03 AE
 7 4.59313E-07 2.18615E-03 AE
 7 2.00152E-05 1.33271E-02 AE
 7 7.12835E-06 1.53342E-02 AE
 7 6.09897E-07 1.73432E-02 AE
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 6 1.09929E-05 1.75819E+00 IE
 6 1.94267E-06 1.76037E+00 IE
 6 3.36470E-07 1.76070E+00 IE
 6 1.71943E-12 2.16828E+00 IE
 6 1.86195E-13 2.18373E+00 IE
 6 2.06702E-15 2.18393E+00 IE
 6 2.03097E-15 2.18402E+00 IE
 6 3.38976E-14 2.18591E+00 IE
 6 5.66817E-15 2.18624E+00 IE
Cf-252       2.645y       198
 2 1.49477E+00 7.76505E-05  X
 2 1.54927E-05 1.69418E-04  X
 2 2.73159E-04 5.77821E-04  X
 2 3.04416E-08 7.77997E-04  X
 2 1.05566E-02 3.66090E-03  X
 2 3.57669E-06 4.78199E-03  X
 2 2.96714E-05 5.56000E-03  X
 2 1.54337E-03 1.26809E-02  X
 2 2.01735E-05 1.30484E-02  X
 2 1.84929E-05 1.41650E-02  X
 2 2.37806E-03 1.47568E-02  X
 1 2.66526E-03 1.48000E-02 PG
 2 2.08769E-02 1.49784E-02  X
 2 3.96851E-04 1.73300E-02  X
 2 6.66127E-04 1.74629E-02  X
 2 4.40542E-06 1.75008E-02  X
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2 2.32139E-02 1.95388E-02  X
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2 7.67997E-06 2.03168E-02  X
2 1.14172E-05 2.05384E-02  X
2 1.83839E-04 2.21119E-02  X
2 1.15497E-05 2.25816E-02  X
2 5.45256E-03 2.28564E-02  X
2 4.15752E-05 2.30608E-02  X
2 3.80504E-05 2.32738E-02  X
2 3.77749E-05 2.33596E-02  X
2 4.91895E-05 2.33917E-02  X
2 2.96958E-06 2.35364E-02  X
2 1.36017E-06 2.36344E-02  X
2 1.13173E-03 2.36395E-02  X
2 2.18802E-06 2.36886E-02  X
2 1.01996E-05 2.37153E-02  X
2 4.53346E-07 2.37490E-02  X
2 2.60811E-06 2.37625E-02  X
2 8.67182E-07 2.37698E-02  X
2 1.05637E-05 2.42392E-02  X
2 9.77461E-06 2.43144E-02  X
2 2.62207E-07 2.44176E-02  X
2 4.26584E-07 2.44288E-02  X
2 1.82419E-06 2.45143E-02  X
2 1.51714E-06 2.45269E-02  X
1 3.66663E-03 3.00000E-02 DG
1 6.95164E-03 3.14000E-02 PG
1 1.48269E-04 4.33990E-02  G
1 2.01513E-04 4.53000E-02 PG
1 3.66663E-03 5.00000E-02 DG
1 2.87072E-04 5.53000E-02 PG
1 3.66663E-03 7.00000E-02 DG
1 5.85401E-04 7.53000E-02 PG
1 8.30256E-04 8.53000E-02 PG
1 3.66663E-03 9.00000E-02 DG
1 1.18487E-03 9.53000E-02 PG
1 1.25980E-04 1.00200E-01  G
2 2.70382E-07 1.05084E-01  X
2 4.23178E-07 1.09866E-01  X
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1 3.94020E-03 1.10000E-01 PG
2 4.93261E-08 1.22914E-01  X
2 9.71035E-08 1.24031E-01  X
2 1.89009E-09 1.24623E-01  X
2 2.09918E-09 1.24845E-01  X
2 1.25065E-08 1.27367E-01  X
2 2.57108E-08 1.27666E-01  X
2 5.74902E-10 1.27940E-01  X
2 6.42096E-10 1.27995E-01  X
2 3.14406E-09 1.28545E-01  X
2 6.39050E-09 1.28620E-01  X
2 1.26237E-10 1.28724E-01  X
2 1.39687E-10 1.28735E-01  X
2 5.40562E-10 1.28820E-01  X
2 9.84974E-10 1.28833E-01  X
1 3.66663E-03 1.30000E-01 DG
1 3.94020E-03 1.30000E-01 PG
1 3.66663E-03 1.50000E-01 DG
1 3.94020E-03 1.50000E-01 PG
1 5.03922E-06 1.54500E-01  G
1 3.94020E-03 1.70000E-01 PG
1 3.66663E-03 1.70000E-01 DG
1 3.94020E-03 1.90000E-01 PG
1 3.66663E-03 1.90000E-01 DG
1 3.09350E-07 2.06900E-01  G
1 9.87864E-03 2.25000E-01 PG
1 8.99991E-03 2.25000E-01 DG
1 9.87864E-03 2.75000E-01 PG
1 8.99991E-03 2.75000E-01 DG
1 9.87864E-03 3.25000E-01 PG
1 8.99991E-03 3.25000E-01 DG
1 9.87864E-03 3.75000E-01 PG
1 8.99991E-03 3.75000E-01 DG
1 9.87864E-03 4.25000E-01 PG
1 2.53331E-02 4.40000E-01 DG
1 9.87864E-03 4.75000E-01 PG
1 9.87864E-03 5.25000E-01 PG
1 1.96665E-02 5.40000E-01 DG
1 9.87864E-03 5.75000E-01 PG
1 1.53332E-02 6.40000E-01 DG
1 1.90255E-02 6.49000E-01 PG
1 1.19999E-02 7.40000E-01 DG
1 1.59015E-02 7.49000E-01 PG
1 9.33324E-03 8.40000E-01 DG
1 1.33404E-02 8.49000E-01 PG
1 1.11451E-02 9.49000E-01 PG
1 9.34390E-03 1.04900E+00 PG
1 3.93329E-02 1.10000E+00 DG
1 7.79596E-03 1.14900E+00 PG
1 6.52947E-03 1.25000E+00 PG
1 5.45999E-03 1.35000E+00 PG
1 4.55937E-03 1.45000E+00 PG
```

```
1 2.39997E-02 1.56000E+00 DG
1 1.61830E-02 1.73000E+00 PG
1 1.26665E-02 2.00000E+00 DG
1 9.40019E-03 2.23000E+00 PG
1 9.33324E-03 2.40000E+00 DG
1 5.43185E-03 2.73000E+00 PG
1 5.33328E-03 2.80000E+00 DG
1 3.15216E-03 3.23000E+00 PG
1 3.99996E-03 3.25000E+00 DG
1 1.82938E-03 3.73000E+00 PG
1 2.86664E-03 3.75000E+00 DG
1 1.06104E-03 4.23000E+00 PG
1 1.39999E-03 4.25000E+00 DG
1 6.13545E-04 4.73000E+00 PG
1 6.66660E-04 4.75000E+00 DG
1 3.57432E-04 5.23000E+00 PG
1 3.16663E-04 5.25000E+00 DG
1 3.26474E-04 5.91000E+00 PG
1 1.09763E-04 6.91000E+00 PG
1 3.65876E-05 7.91000E+00 PG
1 1.23835E-05 8.91000E+00 PG
1 4.22164E-06 9.91000E+00 PG
5 1.92239E-01 1.27952E+00 DB
7 4.22987E-01 9.71089E-05 AE
7 2.26405E-01 1.90616E-04 AE
7 2.32371E-03 1.93098E-04 AE
7 3.92271E-01 2.79669E-04 AE
7 6.48319E-02 6.01166E-04 AE
7 1.36037E-02 1.79582E-03 AE
7 1.26834E-01 2.67813E-03 AE
7 2.81010E-02 3.27252E-03 AE
7 1.76939E-03 3.98173E-03 AE
7 3.09344E-02 1.12900E-02 AE
7 1.63731E-02 1.50636E-02 AE
6 2.22825E-03 1.88450E-02 IE
7 2.03517E-03 1.88812E-02 AE
6 5.84886E-02 1.96230E-02 IE
6 4.95228E-02 2.44050E-02 IE
6 8.79976E-07 2.56400E-02 IE
6 3.06661E-02 3.85700E-02 IE
6 1.14428E-02 4.33990E-02 IE
6 5.63966E-05 7.56460E-02 IE
6 1.01380E-03 7.64240E-02 IE
6 4.31402E-08 7.80400E-02 IE
6 6.43635E-04 8.12060E-02 IE
7 1.70654E-08 8.29002E-02 AE
6 4.81693E-04 9.53710E-02 IE
6 1.80982E-04 1.00200E-01 IE
7 9.90826E-09 1.01252E-01 AE
7 1.40292E-09 1.19627E-01 AE
6 5.50383E-07 1.29946E-01 IE
6 5.93595E-06 1.30724E-01 IE
6 3.08900E-06 1.35506E-01 IE
```

```
 6 2.67012E-06 1.49671E-01 IE
 6 1.00762E-06 1.54500E-01 IE
 6 1.52783E-08 1.82346E-01 IE
 6 1.04155E-07 1.83124E-01 IE
 6 4.61017E-08 1.87906E-01 IE
 6 4.60365E-08 2.02071E-01 IE
 6 1.73863E-08 2.06900E-01 IE
 8 5.80623E-07 5.62115E+00  A
 8 1.93541E-05 5.82476E+00  A
 8 2.32249E-03 5.97681E+00  A
 8 1.51930E-01 6.07542E+00  A
 8 8.14808E-01 6.11813E+00  A
 9 5.80623E-07 9.07226E-02 AR
 9 1.93541E-05 9.40088E-02 AR
 9 2.32249E-03 9.64628E-02 AR
 9 1.51930E-01 9.80543E-02 AR
 9 8.14808E-01 9.87436E-02 AR
10 6.18400E-02 9.40435E+01 FF
11 1.16414E-01 2.30603E+00  N
F-18        109.77m         2
 3 1.93460E+00 5.11000E-01 AQ
 4 9.67300E-01 2.49776E-01 B+
================================================
```

Emitted radiation types are:

X = X ray

G = Gamma ray

PG = Prompt Gamma Ray

DG = Delayed Gamma Ray

AQ = Annihilation Quantum -- an annihilation photon from antiparticle annihilation.

B- = Negative Beta particle

B+ = Positive Beta particle

AE = Auger electron

IE = Internal conversion electron

A = Alpha particle

AR = Recoil atom from alpha particle emission.

FF = Fission Fragment

All of X, G, DG, PG and AQ denote an IONISING PHOTON with a line energy distribution. Whether a photon is X, G, DG or PG, is irrelevant from the viewpoint of the transport of the radiation in MCNP.

Both an IE and a AE denotes a LINE ENERGY ELECTRON . Whether an electron is an IE or a AE is irrelevant from the viewpoint of MCNP. Both are sharply defined line energy electron emissions.

An A denotes an alpha particle with a sharp, line energy emission spectrum.

An AR denotes the heavy recoiling ion resulting from Alpha particle emission. Energies are also sharp lines, because these are two-particle systems.

A B- denotes a negative beta electron emission. The energy spread is continuous, because these are 3-particle systems.

A B+ denotes a positive beta electron emission. The energy spread is continuous, because these are 3-particle systems. Upon coming to rest, this anti-particle will annihilate.

Look at the column structure of the **ICRP-07.RAD** database:

```
2 8.38497E+00 6.06280E-05  X
```

Column 1 = an integer emission type designator. The mapping can easily be decoded.
Column 2 = Emission Yield Y.
Column 3 = Emission line energy (X, G, AQ, DG, PG, A, AR) or E_average (B-, B+, FF).
Column 4 = Mnemonic for particle designation, mapping to the integer in column 1

The format of the second ICRP-107 database (circa 2 MByte), namely BETA particle emission, is as follows:

```
Ac-226        112
0.00000 1.486E+00
0.00010 1.486E+00
0.00011 1.486E+00
0.00012 1.486E+00
0.00013 1.486E+00
0.00014 1.486E+00
0.00015 1.486E+00
0.00016 1.486E+00
0.00018 1.486E+00
0.00020 1.486E+00
0.00022 1.486E+00
0.00024 1.486E+00
0.00026 1.486E+00
0.00028 1.486E+00
0.00030 1.486E+00
0.00032 1.486E+00
0.00036 1.486E+00
0.00040 1.486E+00
0.00045 1.486E+00
0.00050 1.486E+00
0.00055 1.486E+00
0.00060 1.486E+00
0.00065 1.486E+00
0.00070 1.486E+00
0.00075 1.486E+00
0.00080 1.486E+00
0.00085 1.486E+00
0.00090 1.486E+00
0.00100 1.486E+00
0.00110 1.485E+00
0.00120 1.485E+00
0.00130 1.485E+00
0.00140 1.485E+00
0.00150 1.485E+00
0.00160 1.485E+00
0.00180 1.485E+00
0.00200 1.485E+00
0.00220 1.485E+00
0.00240 1.485E+00
```

0.00260 1.485E+00
0.00280 1.485E+00
0.00300 1.485E+00
0.00320 1.485E+00
0.00360 1.485E+00
0.00400 1.485E+00
0.00450 1.485E+00
0.00500 1.485E+00
0.00550 1.484E+00
0.00600 1.484E+00
0.00650 1.484E+00
0.00700 1.484E+00
0.00750 1.484E+00
0.00800 1.484E+00
0.00850 1.484E+00
0.00900 1.483E+00
0.01000 1.483E+00
0.01100 1.483E+00
0.01200 1.482E+00
0.01300 1.482E+00
0.01400 1.482E+00
0.01500 1.481E+00
0.01600 1.481E+00
0.01800 1.480E+00
0.02000 1.483E+00
0.02200 1.485E+00
0.02400 1.486E+00
0.02600 1.488E+00
0.02800 1.489E+00
0.03000 1.491E+00
0.03200 1.492E+00
0.03600 1.495E+00
0.04000 1.497E+00
0.04500 1.500E+00
0.05000 1.502E+00
0.05500 1.504E+00
0.06000 1.506E+00
0.06500 1.508E+00
0.07000 1.509E+00
0.07500 1.511E+00
0.08000 1.512E+00
0.08500 1.512E+00
0.09000 1.513E+00
0.10000 1.513E+00
0.11000 1.512E+00
0.12000 1.510E+00
0.13000 1.507E+00
0.14000 1.503E+00
0.15000 1.498E+00
0.16000 1.492E+00
0.18000 1.478E+00
0.20000 1.461E+00
0.22000 1.440E+00

0.24000 1.416E+00
0.26000 1.389E+00
0.28000 1.359E+00
0.30000 1.326E+00
0.32000 1.291E+00
0.36000 1.213E+00
0.40000 1.126E+00
0.45000 1.007E+00
0.50000 8.802E-01
0.55000 7.491E-01
0.60000 6.176E-01
0.65000 4.899E-01
0.70000 3.704E-01
0.75000 2.637E-01
0.80000 1.745E-01
0.85000 1.078E-01
0.90000 6.770E-02
1.00000 1.893E-02
1.10000 4.064E-04
1.11700 0.000E+00
Ac-227 73
0.00000 7.974E+01
0.00010 7.925E+01
0.00011 7.921E+01
0.00012 7.916E+01
0.00013 7.911E+01
0.00014 7.906E+01
0.00015 7.901E+01
0.00016 7.896E+01
0.00018 7.887E+01
0.00020 7.877E+01
0.00022 7.868E+01
0.00024 7.858E+01
0.00026 7.849E+01
0.00028 7.839E+01
0.00030 7.829E+01
0.00032 7.820E+01
0.00036 7.801E+01
0.00040 7.782E+01
0.00045 7.758E+01
0.00050 7.734E+01
0.00055 7.710E+01
0.00060 7.687E+01
0.00065 7.663E+01
0.00070 7.640E+01
0.00075 7.616E+01
0.00080 7.593E+01
0.00085 7.569E+01
0.00090 7.546E+01
0.00100 7.499E+01
0.00110 7.453E+01
0.00120 7.407E+01
0.00130 7.361E+01

```
0.00140 7.315E+01
0.00150 7.269E+01
0.00160 7.224E+01
0.00180 7.133E+01
0.00200 7.044E+01
0.00220 6.955E+01
0.00240 6.867E+01
0.00260 6.779E+01
0.00280 6.693E+01
0.00300 6.607E+01
0.00320 6.522E+01
0.00360 6.354E+01
0.00400 6.189E+01
0.00450 5.987E+01
0.00500 5.790E+01
0.00550 5.598E+01
0.00600 5.410E+01
0.00650 5.228E+01
0.00700 5.050E+01
0.00750 4.876E+01
0.00800 4.705E+01
0.00850 4.538E+01
0.00900 4.375E+01
0.01000 4.059E+01
0.01100 3.756E+01
0.01200 3.469E+01
0.01300 3.195E+01
0.01400 2.936E+01
0.01500 2.692E+01
0.01600 2.462E+01
0.01800 2.046E+01
0.02000 1.694E+01
0.02200 1.390E+01
0.02400 1.116E+01
0.02600 8.736E+00
0.02800 6.628E+00
0.03000 4.843E+00
0.03200 3.386E+00
0.03600 1.473E+00
0.04000 4.430E-01
0.04480 0.000E+00

==============================
```

The Beta-particle emission database gives a probability density function in E-Y format.

Unlike the **ICRP-07.RAD** line emission database, which is in Y-E format, the .BET database is already in the desired E-Y format. For a probability density function, Energy must be monotonically increasing in MCNP.

The MCNP format into which the **ICRP-07.RAD** and the **ICRP-07.BET** databases must be re-cast by the requested code, is illustrated by the following general format (Co-60 is used as example). Information that must be hard-coded is shown in RED and BLUE. Information calculated from the **ICRP-107.RAD** and **ICRP-107.BET** databases are shown in black.

Note that MCNP receives the E-Y line emission spectrum data in {E-Y} = {SI-SP} format, i.e. ICRP-107 Y-E columns in the .RAD database must be swapped. In MCNP the Energy column is the SI column, and the emission Yield column is the SP column.

MCNP5 and MCNPX could only transport photons and electrons having E >= 1E-3 MeV. MCNP6 can transport down to E = 1E-5 MeV. There must be an input specification of E_min for photons and electrons. If E_min = 1E-3 MeV is selected for a particle type, emission LINE energies falling below E_min, must be left out, ignored.

Please sort all the line energy emission EY matrixes by E as default, from high E to low E.

The LHS indentations seen below, must be 6 or 7 spaces, to serve as card continuation indicators to MCNP.

Note that # denotes a generic heavy ion (A > 4).

```
c
c == GENERAL SDEF FOR [enter NUCLIDE NAME]
c
mode   N  P  E  A  #
c
c Cutoff Energies for Charged Particle Transport:
cut:E      J  1E-3  J  J     $ Cutoff energy for electron       transport
cut:A      J  1E-3  J  J     $ Cutoff energy for alpha-particle transport
cut:#      J  1E-3  J  J     $ Cutoff energy for heavy ion      transport
c
SDEF   POS = 0  0  0
       PAR = D7
       ERG = FPAR  = D10
c
#       SI7      SP7              DS10
        L        D                S
        N        0.0000           21          $ Neutrons
        P        1.9985949        22          $ Photons
        E        4.4766187E-04    23          $ Line-Electrons
        E        1.0000000        24          $ Beta-Electrons (-)
       -E        0.0000           25          $ Beta-Positrons (+)
        A        0.0000           26          $ Alpha-Particles
        90232    0.0000           27          $ Recoil atom from alpha emission
        59149    0.0000           28          $ Typical heavy fission fragment
        40100    0.0000           29          $ Typical light fission fragment
c
c [enter NUCLIDE NAME] Ionising Photons
#      SI22              SP22
       L                 D
       2.505690E+00      2.000000E-08
       2.158570E+00      1.200000E-05
       1.332490E+00      9.998260E-01
       1.173230E+00      9.985000E-01
       8.261000E-01      7.600000E-05
       3.471400E-01      7.500000E-05
       8.288160E-03      4.867420E-09
       8.287840E-03      3.349000E-09
       8.224590E-03      7.660650E-06
       8.222290E-03      3.906860E-06
       7.435780E-03      6.238820E-05
       7.417820E-03      3.194630E-05
c
c
c [enter NUCLIDE NAME] Line-Electrons
#      SI23                SP23
       L                   D
       2.5048200E+00       8.7611000E-16
       2.5048100E+00       1.4936100E-15
       2.5046900E+00       1.5461500E-13
       2.4973900E+00       1.5580700E-12
       2.1577000E+00       3.5967500E-13
       2.1576900E+00       2.3956200E-13
       2.1575600E+00       5.1796200E-11
```

```
       2.1502700E+00        5.3678900E-10
       1.3316200E+00        1.0630400E-07
       1.3316100E+00        8.8254700E-08
       1.3314900E+00        1.0889700E-05
       1.3241900E+00        1.1353700E-04
       1.1723600E+00        1.5670000E-07
       1.1723400E+00        1.3511000E-07
       1.1722200E+00        1.4379800E-05
       1.1649300E+00        1.4995000E-04
       8.2523300E-01        2.3986400E-11
       8.2521500E-01        2.5422500E-11
       8.2509400E-01        2.2261900E-09
       8.1779700E-01        2.3140400E-08
       3.4627300E-01        1.3871300E-09
       3.4625500E-01        1.2041100E-09
       3.4613400E-01        3.5556100E-08
       3.3883700E-01        3.7687600E-07
       8.1334900E-03        2.0573700E-06
       7.3136200E-03        3.1191600E-05
       6.4978800E-03        1.2472900E-04
c
c [enter NUCLIDE NAME] Negative Beta-Electrons (-)
#      SI24                 SP24
       A                    D
       0.00000              6.626E+00
       0.00010              6.623E+00
       0.00011              6.623E+00
       0.00012              6.622E+00
       0.00013              6.622E+00
       0.00014              6.622E+00
       0.00015              6.621E+00
       0.00016              6.621E+00
       0.00018              6.621E+00
       0.00020              6.620E+00
       0.00022              6.619E+00
       0.00024              6.619E+00
       0.00026              6.618E+00
       0.00028              6.618E+00
       0.00030              6.617E+00
       0.00032              6.616E+00
       0.00036              6.615E+00
       0.00040              6.614E+00
       0.00045              6.613E+00
       0.00050              6.611E+00
       0.00055              6.610E+00
       0.00060              6.608E+00
       0.00065              6.607E+00
       0.00070              6.606E+00
       0.00075              6.604E+00
       0.00080              6.603E+00
       0.00085              6.601E+00
       0.00090              6.600E+00
       0.00100              6.597E+00
```

0.00110	6.594E+00
0.00120	6.591E+00
0.00130	6.588E+00
0.00140	6.586E+00
0.00150	6.583E+00
0.00160	6.580E+00
0.00180	6.574E+00
0.00200	6.568E+00
0.00220	6.563E+00
0.00240	6.557E+00
0.00260	6.551E+00
0.00280	6.545E+00
0.00300	6.540E+00
0.00320	6.534E+00
0.00360	6.523E+00
0.00400	6.516E+00
0.00450	6.507E+00
0.00500	6.497E+00
0.00550	6.488E+00
0.00600	6.478E+00
0.00650	6.469E+00
0.00700	6.459E+00
0.00750	6.450E+00
0.00800	6.440E+00
0.00850	6.431E+00
0.00900	6.422E+00
0.01000	6.403E+00
0.01100	6.385E+00
0.01200	6.367E+00
0.01300	6.349E+00
0.01400	6.332E+00
0.01500	6.315E+00
0.01600	6.298E+00
0.01800	6.265E+00
0.02000	6.234E+00
0.02200	6.203E+00
0.02400	6.172E+00
0.02600	6.142E+00
0.02800	6.113E+00
0.03000	6.083E+00
0.03200	6.053E+00
0.03600	5.993E+00
0.04000	5.932E+00
0.04500	5.853E+00
0.05000	5.771E+00
0.05500	5.685E+00
0.06000	5.596E+00
0.06500	5.502E+00
0.07000	5.405E+00
0.07500	5.304E+00
0.08000	5.198E+00
0.08500	5.089E+00
0.09000	4.977E+00

```
        0.10000             4.742E+00
        0.11000             4.494E+00
        0.12000             4.235E+00
        0.13000             3.967E+00
        0.14000             3.692E+00
        0.15000             3.411E+00
        0.16000             3.127E+00
        0.18000             2.557E+00
        0.20000             1.998E+00
        0.22000             1.469E+00
        0.24000             9.907E-01
        0.26000             5.822E-01
        0.28000             2.658E-01
        0.30000             6.436E-02
        0.32000             1.027E-03
        0.36000             1.050E-03
        0.40000             1.070E-03
        0.45000             1.089E-03
        0.50000             1.102E-03
        0.55000             1.108E-03
        0.60000             1.108E-03
        0.65000             1.102E-03
        0.70000             1.089E-03
        0.75000             1.070E-03
        0.80000             1.044E-03
        0.85000             1.010E-03
        0.90000             9.695E-04
        1.00000             8.627E-04
        1.10000             7.174E-04
        1.20000             5.296E-04
        1.30000             3.083E-04
        1.40000             9.574E-05
        1.49139             0.000E+00
        1.50000             0.000E+00
c
c === END SDEF [enter NUCLIDE NAME] ALL PARTICLES ===
c
```

Note the illustrative EXAMPLE heavy particle specifier **90232** above (this has nothing to do with Co-60). The # is a generic specifier that heavy ion transport is needed, but the heavy particle specifier must be a specific ZAID in the SI7 column. It is calculated as follows:
ZAID = ZA ID designator = 1000Z + A.
Z = (Z_parent - 2) because 2 protons are emitted in an alpha particle.
A = (A - 4) because 4 nucleons are emitted in the alpha-particle.

Example: If the alpha particle emitter is U-238, the Z and A of the # for the recoil atom must be Z = 92 - 2 = 90 and A = 238 - 4 = 234 so that ZAID of the # for alpha-emission recoil must be specified as 90234 in the SI7 column.

=====

The fission fragment ZAID specifications are calculated as follows, using spontaneous fission of Cf-252 as an example:

$\nu := 3.77$

$Z_0 := 98$ $\quad A_0 := 252$

$Z_1 := \text{ceil}(Z_0 \cdot 0.4)$ $\quad Z_2 := \text{ceil}(Z_0 \cdot 0.6)$

$A_1 := \text{ceil}[(A_0 - \text{ceil}(\nu)) \cdot 0.4]$ $\quad A_2 := \text{ceil}[(A_0 - \text{ceil}(\nu)) \cdot 0.6]$

$1000 \cdot Z_{.1} + A_{.1} = 40100$

$1000 Z_2 + A_2 = 59149$

(The Roundoff function instead of the ceiling may be even better.)

=====

Next, we look at the calculated entries in the SP7 column:

```
#       SI7       SP7               DS10
        L         D                 S
        N         0.0000            21          $ Neutrons
        P         1.9985949         22          $ Photons
        E         4.4766187E-04     23          $ Line-Electrons
        E         1.0000000         24          $ Beta-Electrons (-)
       -E         0.0000            25          $ Beta-Positrons (+)
        A         0.0000            26          $ Alpha-Particles
        90232     0.0000            27          $ Recoil atom from alpha emission
        59149     0.0000            28          $ Typical light fission fragment
        40100     0.0000            29          $ Typical heavy fission fragment
```

Example from Co-60 emissions:

1. The total photon yield FOR EMITTED PHOTONS WITH E >= 1E-3 MeV is 1.9985949. This is the sum of the Y column in the EY matrix for line photon emission above E = 1E-3 MeV.
2. The total discrete-line electron ("LE") yield FOR EMITTED LINE ELECTRONS (AE & IE) having E >= E_cut_electron, is 4.4766187E-04. This is the sum of the Y column in the EY matrix for line electron emission above E = E_cut_electron.
3. The total NEGATIVE BETA electron yield FOR EMITTED BETA ELECTRONS having E >= E_cut_electron, is 1.0000. This information is the sum of the Yields of all BETA emissions for this radionuclide, in the **ICRP-107.RAD** database. This information is NOT in the **ICRP-107.BET** database, but from the **ICRP-107.RAD** database.

Note that nuclides such as F-18 emits a POSITRON, and that a positron is specified as -E in the SI 7 column.

Important: The **ICRP-107.RAD** database also includes AQ = Annihilation Quantum Photon emission information. Thus must always be ***ignored,*** because the version of MCNP used in Necsa RRT, models antiparticle transport in detail, and will internally and automatically generate the AQ = Annihilation Quantum Photon once the positron energy reaches E = E_cut_electron (default = 1E-3 MeV). (Only users who use MCNP in "Mickey-Mouse mode" such as "MODE P" will need explicit AQ specifications. This is generally not done in RRT. An input selection may be appropriate.)

Important: The **ICRP-107.RAD** database contains the average neutron emission energy and total neutron emission yield for spontaneous fission nuclides such as Cf-252. These must usually be ignored, because WATT fission neutron energy spectrum

parameters are specified in MCNP -- SF neutron information from ICRP-107 is irrelevant for the MCNP user. The requested code must print a prompt such as [INSERT WATT SPECTRUM PARAMETERS HERE].

Goal 2: Spectroscopic SDEF Consolidator for a Radionuclide Cocktail

Burnup, Activation & Fission codes such as ORIGIN-S, FISSPACT-II and CINDER-90 print a list of nuclides produced by activation and fission in one of the following formats:

Nuclide	Activity (Ci)
Co-60	1.798654E2
Cs-137	3.376574E3
...	...

or

no.	zaid	mass (gm)	activity (Ci)	sp. act. (Ci/gm)	atom den. (a/b-cm)	atom fr.	mass fr.
1	8016	1.478E+05	0.000E+00	0.000E+00	3.615E-02	3.350E-01	8.729E-01
2	1001	1.848E+04	0.000E+00	0.000E+00	7.172E-02	6.645E-01	1.091E-01
3	54134	1.738E-06	0.000E+00	0.000E+00	5.078E-14	4.706E-13	1.027E-11
4	56138	1.541E-06	0.000E+00	0.000E+00	4.371E-14	4.050E-13	9.100E-12
5	57139	1.471E-06	0.000E+00	0.000E+00	4.144E-14	3.840E-13	8.689E-12
6	58140	1.436E-06	0.000E+00	0.000E+00	4.016E-14	3.721E-13	8.481E-12
7	54136	1.416E-06	0.000E+00	0.000E+00	4.077E-14	3.778E-13	8.365E-12
8	55135	1.416E-06	1.632E-09	1.152E-03	4.106E-14	3.805E-13	8.363E-12
9	60143	1.405E-06	0.000E+00	0.000E+00	3.847E-14	3.565E-13	8.300E-12
10	58142	1.370E-06	6.908E-20	5.042E-14	3.777E-14	3.500E-13	8.092E-12
11	55137	1.366E-06	1.189E-04	8.704E+01	3.903E-14	3.616E-13	8.066E-12
12	59141	1.360E-06	0.000E+00	0.000E+00	3.775E-14	3.498E-13	8.030E-12
13	42100	1.038E-06	0.000E+00	0.000E+00	4.066E-14	3.768E-13	6.133E-12
14	40096	1.004E-06	0.000E+00	0.000E+00	4.093E-14	3.793E-13	5.927E-12
15	40094	1.003E-06	0.000E+00	0.000E+00	4.180E-14	3.874E-13	5.926E-12
16	43099	9.907E-07	1.696E-08	1.712E-02	3.919E-14	3.631E-13	5.851E-12
17	42095	9.771E-07	0.000E+00	0.000E+00	4.028E-14	3.732E-13	5.770E-12
18	40093	9.678E-07	2.434E-09	2.515E-03	4.075E-14	3.776E-13	5.715E-12
19	60145	9.411E-07	3.870E-20	4.112E-14	2.541E-14	2.354E-13	5.558E-12
20	42097	9.374E-07	0.000E+00	0.000E+00	3.784E-14	3.507E-13	5.536E-12
21	42098	9.290E-07	0.000E+00	0.000E+00	3.712E-14	3.440E-13	5.486E-12
22	40092	9.137E-07	0.000E+00	0.000E+00	3.889E-14	3.604E-13	5.396E-12
23	54132	9.132E-07	0.000E+00	0.000E+00	2.708E-14	2.510E-13	5.393E-12
24	44101	8.634E-07	0.000E+00	0.000E+00	3.347E-14	3.102E-13	5.099E-12
25	40091	8.547E-07	0.000E+00	0.000E+00	3.678E-14	3.408E-13	5.048E-12
26	60144	7.675E-07	9.116E-19	1.188E-12	2.086E-14	1.933E-13	4.532E-12
27	55133	7.641E-07	0.000E+00	0.000E+00	2.249E-14	2.084E-13	4.512E-12
28	60146	7.222E-07	0.000E+00	0.000E+00	1.936E-14	1.794E-13	4.265E-12
29	44102	7.190E-07	0.000E+00	0.000E+00	2.760E-14	2.558E-13	4.246E-12
30	39089	6.901E-07	0.000E+00	0.000E+00	3.036E-14	2.814E-13	4.075E-12

31	54131	6.245E-07	0.000E+00	0.000E+00	1.866E-14	1.729E-13	3.688E-12
32	58144	5.364E-07	1.708E-03	3.185E+03	1.458E-14	1.351E-13	3.168E-12
33	45103	5.124E-07	0.000E+00	0.000E+00	1.948E-14	1.805E-13	3.026E-12
34	38088	5.121E-07	0.000E+00	0.000E+00	2.279E-14	2.112E-13	3.024E-12
35	61147	4.232E-07	3.925E-04	9.276E+02	1.127E-14	1.044E-13	2.499E-12
36	60148	4.085E-07	0.000E+00	0.000E+00	1.080E-14	1.001E-13	2.413E-12
37	52130	3.882E-07	0.000E+00	0.000E+00	1.169E-14	1.083E-13	2.292E-12
38	37087	3.705E-07	3.175E-14	8.570E-08	1.668E-14	1.545E-13	2.188E-12
39	44104	3.226E-07	0.000E+00	0.000E+00	1.215E-14	1.126E-13	1.905E-12
40	36086	2.983E-07	0.000E+00	0.000E+00	1.358E-14	1.259E-13	1.762E-12
41	62149	2.659E-07	3.192E-19	1.200E-12	6.986E-15	6.474E-14	1.571E-12
42	38090	1.974E-07	2.789E-05	1.413E+02	8.591E-15	7.961E-14	1.166E-12
43	46105	1.670E-07	0.000E+00	0.000E+00	6.227E-15	5.770E-14	9.862E-13
44	60150	1.614E-07	0.000E+00	0.000E+00	4.212E-15	3.903E-14	9.533E-13
45	36084	1.354E-07	0.000E+00	0.000E+00	6.311E-15	5.848E-14	7.995E-13
46	37085	1.327E-07	0.000E+00	0.000E+00	6.113E-15	5.664E-14	7.836E-13
47	62147	1.217E-07	2.793E-15	2.296E-08	3.240E-15	3.002E-14	7.185E-13
48	62151	1.035E-07	2.725E-06	2.632E+01	2.684E-15	2.487E-14	6.114E-13
49	36083	7.382E-08	0.000E+00	0.000E+00	3.483E-15	3.227E-14	4.359E-13
50	52128	7.236E-08	0.000E+00	0.000E+00	2.213E-15	2.051E-14	4.273E-13
51	62152	6.670E-08	0.000E+00	0.000E+00	1.718E-15	1.592E-14	3.939E-13
52	53129	5.567E-08	9.833E-12	1.766E-04	1.689E-15	1.565E-14	3.287E-13
53	34082	4.387E-08	1.373E-24	3.130E-17	2.095E-15	1.941E-14	2.591E-13
54	63153	3.996E-08	0.000E+00	0.000E+00	1.022E-15	9.473E-15	2.360E-13
55	36085	3.601E-08	1.414E-05	3.927E+02	1.659E-15	1.537E-14	2.127E-13
56	44106	3.551E-08	1.178E-04	3.318E+03	1.312E-15	1.215E-14	2.097E-13
57	46106	3.467E-08	0.000E+00	0.000E+00	1.281E-15	1.187E-14	2.048E-13
58	53127	3.227E-08	0.000E+00	0.000E+00	9.948E-16	9.218E-15	1.906E-13
59	56137	3.193E-08	0.000E+00	0.000E+00	9.125E-16	8.455E-15	1.886E-13
60	35081	2.721E-08	0.000E+00	0.000E+00	1.315E-15	1.219E-14	1.607E-13
61	46107	2.577E-08	1.326E-11	5.145E-04	9.430E-16	8.738E-15	1.522E-13
62	41095	2.264E-08	8.907E-04	3.934E+04	9.332E-16	8.647E-15	1.337E-13
63	40095	1.954E-08	4.199E-04	2.149E+04	8.054E-16	7.463E-15	1.154E-13
64	62154	1.891E-08	0.000E+00	0.000E+00	4.807E-16	4.454E-15	1.117E-13
65	34080	1.700E-08	0.000E+00	0.000E+00	8.319E-16	7.709E-15	1.004E-13
66	50126	1.167E-08	3.314E-10	2.839E-02	3.626E-16	3.360E-15	6.892E-14
67	39091	1.152E-08	2.829E-04	2.455E+04	4.959E-16	4.595E-15	6.805E-14
68	46108	9.652E-09	0.000E+00	0.000E+00	3.499E-16	3.243E-15	5.700E-14
69	63155	7.086E-09	3.494E-06	4.931E+02	1.789E-16	1.658E-15	4.185E-14
70	34079	5.828E-09	8.000E-10	1.373E-01	2.889E-16	2.677E-15	3.442E-14
71	47109	5.611E-09	0.000E+00	0.000E+00	2.016E-16	1.868E-15	3.314E-14
72	50124	5.475E-09	0.000E+00	0.000E+00	1.729E-16	1.602E-15	3.233E-14
73	51125	5.456E-09	5.720E-06	1.048E+03	1.709E-16	1.583E-15	3.222E-14
74	40090	4.872E-09	0.000E+00	0.000E+00	2.120E-16	1.964E-15	2.877E-14
75	46110	4.656E-09	0.000E+00	0.000E+00	1.657E-16	1.536E-15	2.749E-14
76	38089	4.643E-09	1.349E-04	2.905E+04	2.043E-16	1.893E-15	2.742E-14
77	2004	4.541E-09	0.000E+00	0.000E+00	4.438E-15	4.113E-14	2.682E-14
78	64156	3.823E-09	0.000E+00	0.000E+00	9.593E-17	8.889E-16	2.258E-14
79	48111	3.149E-09	0.000E+00	0.000E+00	1.111E-16	1.029E-15	1.860E-14
80	51123	3.145E-09	0.000E+00	0.000E+00	1.001E-16	9.277E-16	1.858E-14
81	50122	3.116E-09	0.000E+00	0.000E+00	9.999E-17	9.265E-16	1.840E-14
82	34078	2.697E-09	0.000E+00	0.000E+00	1.354E-16	1.255E-15	1.593E-14
83	48113	2.602E-09	8.859E-22	3.405E-13	9.016E-17	8.354E-16	1.537E-14

84	48116	2.538E-09	0.000E+00	0.000E+00	8.568E-17	7.939E-16	1.499E-14
85	50120	2.496E-09	0.000E+00	0.000E+00	8.142E-17	7.545E-16	1.474E-14
86	50117	2.456E-09	0.000E+00	0.000E+00	8.220E-17	7.617E-16	1.451E-14
87	48112	2.404E-09	0.000E+00	0.000E+00	8.406E-17	7.789E-16	1.420E-14
88	49115	2.258E-09	1.593E-20	7.055E-12	7.689E-17	7.125E-16	1.334E-14
89	50118	2.207E-09	0.000E+00	0.000E+00	7.323E-17	6.786E-16	1.303E-14
90	48114	2.068E-09	0.000E+00	0.000E+00	7.104E-17	6.582E-16	1.221E-14
91	64157	1.594E-09	0.000E+00	0.000E+00	3.973E-17	3.681E-16	9.411E-15
92	50119	1.495E-09	0.000E+00	0.000E+00	4.919E-17	4.558E-16	8.830E-15
93	52125	1.482E-09	0.000E+00	0.000E+00	4.641E-17	4.300E-16	8.750E-15
94	51121	1.388E-09	0.000E+00	0.000E+00	4.490E-17	4.160E-16	8.194E-15
95	56136	1.243E-09	0.000E+00	0.000E+00	3.579E-17	3.317E-16	7.343E-15
96	64155	1.131E-09	0.000E+00	0.000E+00	2.856E-17	2.647E-16	6.680E-15
97	34077	1.010E-09	0.000E+00	0.000E+00	5.138E-17	4.761E-16	5.966E-15
98	64158	8.566E-10	0.000E+00	0.000E+00	2.122E-17	1.966E-16	5.059E-15
99	44103	8.150E-10	2.634E-05	3.232E+04	3.098E-17	2.871E-16	4.813E-15
100	63151	7.967E-10	0.000E+00	0.000E+00	2.065E-17	1.914E-16	4.705E-15
101	58141	5.662E-10	1.614E-05	2.851E+04	1.572E-17	1.457E-16	3.344E-15
102	32076	3.885E-10	0.000E+00	0.000E+00	2.002E-17	1.855E-16	2.294E-15
103	52126	3.195E-10	0.000E+00	0.000E+00	9.928E-18	9.199E-17	1.887E-15
104	65159	2.649E-10	0.000E+00	0.000E+00	6.520E-18	6.041E-17	1.564E-15
105	33075	1.314E-10	0.000E+00	0.000E+00	6.859E-18	6.356E-17	7.758E-16
106	50115	1.142E-10	0.000E+00	0.000E+00	3.886E-18	3.601E-17	6.741E-16
107	42096	8.614E-11	0.000E+00	0.000E+00	3.514E-18	3.256E-17	5.087E-16
108	64160	8.422E-11	0.000E+00	0.000E+00	2.060E-18	1.909E-17	4.974E-16
109	39090	5.130E-11	2.790E-05	5.438E+05	2.232E-18	2.068E-17	3.030E-16
110	54130	4.753E-11	0.000E+00	0.000E+00	1.431E-18	1.326E-17	2.807E-16
111	32074	4.078E-11	0.000E+00	0.000E+00	2.158E-18	2.000E-17	2.408E-16
112	44099	3.544E-11	0.000E+00	0.000E+00	1.402E-18	1.299E-17	2.093E-16
113	50123	3.388E-11	2.786E-07	8.223E+03	1.078E-18	9.992E-18	2.001E-16
114	89227	3.010E-11	2.177E-09	7.233E+01	5.186E-19	4.805E-18	1.777E-16
115	66161	2.266E-11	0.000E+00	0.000E+00	5.509E-19	5.104E-18	1.338E-16
116	32073	1.224E-11	0.000E+00	0.000E+00	6.567E-19	6.086E-18	7.230E-17
117	62150	7.428E-12	0.000E+00	0.000E+00	1.938E-19	1.796E-18	4.387E-17
118	36082	7.388E-12	0.000E+00	0.000E+00	3.528E-19	3.270E-18	4.363E-17
119	57138	7.240E-12	1.787E-19	2.469E-08	2.054E-19	1.903E-18	4.276E-17
120	66162	4.252E-12	0.000E+00	0.000E+00	1.027E-19	9.520E-19	2.511E-17
121	32072	3.182E-12	0.000E+00	0.000E+00	1.731E-19	1.604E-18	1.879E-17
122	52124	2.093E-12	0.000E+00	0.000E+00	6.607E-20	6.122E-19	1.236E-17
123	49113	2.073E-12	0.000E+00	0.000E+00	7.182E-20	6.655E-19	1.224E-17
124	66163	1.641E-12	0.000E+00	0.000E+00	3.941E-20	3.652E-19	9.694E-18
125	38086	1.296E-12	0.000E+00	0.000E+00	5.900E-20	5.467E-19	7.652E-18
126	55134	1.218E-12	1.577E-09	1.295E+03	3.558E-20	3.297E-19	7.192E-18
127	31071	9.825E-13	0.000E+00	0.000E+00	5.419E-20	5.022E-19	5.802E-18
128	44100	9.220E-13	0.000E+00	0.000E+00	3.610E-20	3.345E-19	5.445E-18
129	56134	4.866E-13	0.000E+00	0.000E+00	1.422E-20	1.317E-19	2.874E-18
130	66164	4.605E-13	0.000E+00	0.000E+00	1.099E-20	1.018E-19	2.719E-18
131	56135	4.427E-13	0.000E+00	0.000E+00	1.284E-20	1.190E-19	2.614E-18
132	67165	2.583E-13	0.000E+00	0.000E+00	6.126E-21	5.677E-20	1.525E-18
133	54128	2.044E-13	0.000E+00	0.000E+00	6.252E-21	5.794E-20	1.207E-18
134	38087	1.894E-13	0.000E+00	0.000E+00	8.527E-21	7.901E-20	1.119E-18
135	31069	1.789E-13	0.000E+00	0.000E+00	1.016E-20	9.411E-20	1.057E-18
136	35079	1.226E-13	0.000E+00	0.000E+00	6.077E-21	5.631E-20	7.239E-19

137	68166	9.926E-14	0.000E+00	0.000E+00	2.340E-21	2.168E-20	5.862E-19
138	52122	7.701E-14	0.000E+00	0.000E+00	2.471E-21	2.290E-20	4.548E-19
139	68167	6.811E-14	0.000E+00	0.000E+00	1.596E-21	1.479E-20	4.023E-19
140	63154	4.547E-14	1.229E-11	2.703E+02	1.156E-21	1.071E-20	2.686E-19
141	41094	3.820E-14	7.164E-15	1.875E-01	1.591E-21	1.475E-20	2.256E-19
142	51124	2.875E-14	5.033E-10	1.751E+04	9.078E-22	8.412E-21	1.698E-19
143	34076	2.068E-14	0.000E+00	0.000E+00	1.066E-21	9.876E-21	1.222E-19
144	36080	1.792E-14	0.000E+00	0.000E+00	8.771E-22	8.127E-21	1.058E-19
145	68168	1.549E-14	0.000E+00	0.000E+00	3.608E-22	3.343E-21	9.147E-20
146	2003	1.242E-14	0.000E+00	0.000E+00	1.611E-20	1.493E-19	7.335E-20
147	59143	1.235E-14	8.316E-10	6.733E+04	3.381E-22	3.133E-21	7.294E-20
148	41093	9.817E-15	0.000E+00	0.000E+00	4.134E-22	3.830E-21	5.797E-20
149	66160	7.273E-15	0.000E+00	0.000E+00	1.779E-22	1.648E-21	4.295E-20
150	1003	4.076E-15	3.919E-11	9.613E+03	5.287E-21	4.899E-20	2.407E-20
151	64154	3.822E-15	0.000E+00	0.000E+00	9.715E-23	9.002E-22	2.257E-20
152	56140	3.427E-15	2.508E-10	7.318E+04	9.581E-23	8.878E-22	2.024E-20
153	46104	3.345E-15	0.000E+00	0.000E+00	1.260E-22	1.167E-21	1.976E-20
154	62148	3.061E-15	9.338E-28	3.051E-13	8.095E-23	7.501E-22	1.808E-20
155	47107	2.748E-15	0.000E+00	0.000E+00	1.006E-22	9.317E-22	1.623E-20
156	54129	2.397E-15	0.000E+00	0.000E+00	7.274E-23	6.740E-22	1.415E-20
157	50116	1.473E-15	0.000E+00	0.000E+00	4.973E-23	4.608E-22	8.701E-21
158	68170	1.404E-15	0.000E+00	0.000E+00	3.233E-23	2.996E-22	8.294E-21
159	60142	1.085E-15	0.000E+00	0.000E+00	2.990E-23	2.770E-22	6.405E-21
160	6012	5.864E-16	0.000E+00	0.000E+00	1.912E-22	1.772E-21	3.463E-21
161	51126	5.549E-16	4.640E-11	8.363E+04	1.724E-23	1.597E-22	3.277E-21
162	57140	5.192E-16	2.888E-10	5.561E+05	1.452E-23	1.345E-22	3.066E-21
163	56132	3.214E-16	0.000E+00	0.000E+00	9.533E-24	8.833E-23	1.898E-21
164	52123	3.069E-16	7.199E-26	2.346E-10	9.768E-24	9.052E-23	1.812E-21
165	7015	2.981E-16	0.000E+00	0.000E+00	7.775E-23	7.204E-22	1.760E-21
166	63156	2.266E-16	1.249E-11	5.513E+04	5.686E-24	5.269E-23	1.338E-21
167	65160	2.260E-16	2.553E-12	1.129E+04	5.529E-24	5.124E-23	1.335E-21
168	1002	1.847E-16	0.000E+00	0.000E+00	3.588E-22	3.325E-21	1.091E-21
169	58138	1.081E-16	0.000E+00	0.000E+00	3.068E-24	2.843E-23	6.387E-22
170	42094	7.999E-17	0.000E+00	0.000E+00	3.332E-24	3.088E-23	4.724E-22
171	60147	5.295E-17	4.286E-12	8.094E+04	1.410E-24	1.306E-23	3.127E-22
172	63152	4.317E-17	7.622E-15	1.765E+02	1.112E-24	1.030E-23	2.550E-22
173	58139	3.608E-17	2.464E-13	6.830E+03	1.016E-24	9.415E-24	2.131E-22
174	64152	1.702E-17	3.708E-28	2.179E-11	4.382E-25	4.060E-24	1.005E-22
175	54126	1.544E-17	0.000E+00	0.000E+00	4.798E-25	4.446E-24	9.119E-23
176	38084	1.196E-17	0.000E+00	0.000E+00	5.578E-25	5.168E-24	7.065E-23
177	46102	1.067E-17	0.000E+00	0.000E+00	4.096E-25	3.795E-24	6.301E-23
178	55136	5.488E-18	4.007E-13	7.301E+04	1.580E-25	1.464E-24	3.241E-23
179	32070	1.690E-18	0.000E+00	0.000E+00	9.452E-26	8.759E-25	9.978E-24
180	37086	1.623E-18	1.324E-13	8.158E+04	7.390E-26	6.848E-25	9.584E-24
181	34074	6.924E-19	0.000E+00	0.000E+00	3.664E-26	3.396E-25	4.089E-24
182	48110	2.636E-19	0.000E+00	0.000E+00	9.381E-27	8.693E-26	1.556E-24
183	56133	2.136E-19	5.463E-17	2.557E+02	6.288E-27	5.827E-26	1.262E-24
184	54133	1.637E-19	3.067E-14	1.874E+05	4.817E-27	4.464E-26	9.665E-25
185	53131	1.564E-19	1.940E-14	1.241E+05	4.674E-27	4.331E-26	9.237E-25
186	52132	6.668E-20	2.026E-14	3.038E+05	1.978E-27	1.833E-26	3.938E-25
187	42099	5.012E-20	2.408E-14	4.805E+05	1.982E-27	1.837E-26	2.960E-25
188	58143	4.385E-20	2.914E-14	6.645E+05	1.200E-27	1.112E-26	2.590E-25
189	61148	2.956E-20	4.860E-15	1.644E+05	7.819E-28	7.245E-27	1.746E-25

190	66158	2.875E-20	0.000E+00	0.000E+00	7.121E-28	6.599E-27	1.698E-25
191	44098	2.371E-20	0.000E+00	0.000E+00	9.476E-28	8.780E-27	1.400E-25
192	61149	1.240E-20	4.914E-15	3.964E+05	3.256E-28	3.017E-27	7.320E-26
193	50125	1.129E-20	1.224E-15	1.084E+05	3.535E-28	3.276E-27	6.666E-26
194	54135	1.041E-20	2.646E-14	2.541E+06	3.019E-28	2.798E-27	6.148E-26
195	45105	6.949E-21	5.871E-15	8.448E+05	2.591E-28	2.401E-27	4.104E-26
196	53135	6.576E-21	2.325E-14	3.536E+06	1.907E-28	1.767E-27	3.884E-26
197	88226	6.015E-21	5.946E-21	9.885E-01	1.041E-28	9.647E-28	3.552E-26
198	61151	2.064E-21	1.509E-15	7.311E+05	5.350E-29	4.957E-28	1.219E-26
199	47111	1.727E-21	2.729E-16	1.580E+05	6.090E-29	5.643E-28	1.020E-26
200	62153	1.593E-21	7.057E-16	4.429E+05	4.076E-29	3.777E-28	9.410E-27
201	44105	8.726E-22	5.871E-15	6.728E+06	3.254E-29	3.015E-28	5.153E-27
202	36078	6.349E-23	0.000E+00	0.000E+00	3.188E-30	2.954E-29	3.750E-28
203	63157	3.944E-23	5.188E-17	1.316E+06	9.831E-31	9.110E-30	2.329E-28
204	53130	1.468E-24	2.864E-18	1.952E+06	4.420E-32	4.095E-31	8.667E-30
205	33074	1.326E-24	1.317E-19	9.934E+04	7.015E-32	6.500E-31	7.828E-30
206	59142	1.372E-27	1.585E-21	1.155E+06	3.782E-35	3.505E-34	8.102E-33
207	89228	6.664E-28	1.489E-21	2.235E+06	1.143E-35	1.059E-34	3.935E-33
208	50114	5.680E-30	0.000E+00	0.000E+00	1.951E-37	1.808E-36	3.354E-35
209	68164	8.922E-31	0.000E+00	0.000E+00	2.129E-38	1.973E-37	5.269E-36
210	28063	1.106E-31	6.275E-30	5.675E+01	6.873E-39	6.369E-38	6.530E-37
211	48108	2.525E-32	0.000E+00	0.000E+00	9.154E-40	8.483E-39	1.491E-37
212	28062	5.131E-33	0.000E+00	0.000E+00	3.241E-40	3.003E-39	3.030E-38
213	50112	1.814E-33	0.000E+00	0.000E+00	6.343E-41	5.878E-40	1.072E-38
214	29063	8.607E-34	0.000E+00	0.000E+00	5.350E-41	4.958E-40	5.083E-39
215	64153	5.575E-34	1.970E-30	3.534E+03	1.426E-41	1.322E-40	3.292E-39
216	28061	2.771E-34	0.000E+00	0.000E+00	1.779E-41	1.649E-40	1.637E-39
217	26057	1.848E-34	0.000E+00	0.000E+00	1.269E-41	1.176E-40	1.091E-39
218	8018	1.848E-34	0.000E+00	0.000E+00	4.015E-41	3.721E-40	1.091E-39
219	26058	1.847E-34	0.000E+00	0.000E+00	1.248E-41	1.156E-40	1.091E-39
220	27059	1.844E-34	0.000E+00	0.000E+00	1.224E-41	1.135E-40	1.089E-39
221	7014	9.239E-35	0.000E+00	0.000E+00	2.581E-41	2.392E-40	5.456E-40
222	28060	9.238E-35	0.000E+00	0.000E+00	6.030E-42	5.588E-41	5.456E-40
223	9019	9.238E-35	0.000E+00	0.000E+00	1.902E-41	1.763E-40	5.456E-40
224	6013	9.238E-35	0.000E+00	0.000E+00	2.779E-41	2.575E-40	5.456E-40
225	10020	9.238E-35	0.000E+00	0.000E+00	1.808E-41	1.675E-40	5.456E-40
226	42092	9.238E-35	0.000E+00	0.000E+00	3.932E-42	3.644E-41	5.456E-40
227	8017	9.238E-35	0.000E+00	0.000E+00	2.126E-41	1.970E-40	5.456E-40
228	44096	9.238E-35	0.000E+00	0.000E+00	3.768E-42	3.492E-41	5.456E-40
229	48106	9.238E-35	0.000E+00	0.000E+00	3.412E-42	3.162E-41	5.456E-40
230	62144	9.238E-35	0.000E+00	0.000E+00	2.511E-42	2.327E-41	5.456E-40
231	52120	9.238E-35	0.000E+00	0.000E+00	3.014E-42	2.793E-41	5.456E-40
232	58136	9.238E-35	0.000E+00	0.000E+00	2.659E-42	2.464E-41	5.456E-40
233	56130	9.238E-35	0.000E+00	0.000E+00	2.782E-42	2.578E-41	5.456E-40
234	68162	9.238E-35	0.000E+00	0.000E+00	2.232E-42	2.068E-41	5.456E-40
235	66156	9.238E-35	2.118E-49	2.293E-15	2.318E-42	2.148E-41	5.456E-40
236	54124	9.238E-35	0.000E+00	0.000E+00	2.917E-42	2.703E-41	5.456E-40
237	26060	9.238E-35	3.674E-37	3.977E-03	6.030E-42	5.587E-41	5.456E-40
238	6014	9.237E-35	4.115E-34	4.455E+00	2.580E-41	2.391E-40	5.455E-40
239	50113	1.963E-35	1.973E-31	1.005E+04	6.802E-43	6.303E-42	1.159E-40
240	26059	3.123E-37	1.555E-32	4.979E+04	2.073E-44	1.921E-43	1.844E-42
241	27060	4.073E-41	4.609E-38	1.132E+03	2.659E-48	2.464E-47	2.405E-46
242	5011	9.239E-43	0.000E+00	0.000E+00	3.283E-49	3.042E-48	5.456E-48

243	3007	9.238E-43	0.000E+00	0.000E+00	5.151E-49	4.773E-48	5.456E-48
244	4009	9.238E-43	0.000E+00	0.000E+00	4.010E-49	3.716E-48	5.456E-48
245	3006	9.238E-43	0.000E+00	0.000E+00	6.008E-49	5.567E-48	5.456E-48
246	5010	9.238E-43	0.000E+00	0.000E+00	3.609E-49	3.344E-48	5.456E-48
	totals	1.663E+05	4.191E-03	2.520E-08	1.079E-01	9.995E-01	9.820E-01

In the last example (ORIGEN), only columns 2 and 4 are used. I will extract the two relevant columns manually, sort by activity and delete the activities that are below concern. The format may be either Name-A (e.g. Cs-137) or ZAID (e.g. 26055 for Fe-55). I will export the consolidated {Nuclide_Designator, Activity} matrix as a space delimited ASCII text file.

The requested code must read these columns of the {Nuclide, Activity} array describing the content of the radionuclide cocktail. Activity unit must be selectable -- Bq, MBq, GBq, TBq, mCi or Ci. These and other options such as CUT values must be set in a well documented .DEFAULTS file, and should not be selected via interactive interrogation.

Code 2 must read the {Nuclide, Activity} array from an ASCII file, and then use the spectroscopic radionuclide emission information obtained from the ICRP database to create one consolidated MCNP input for the sum of all the emissions from the cocktail of radionuclides. This will be a spectroscopic source definition for all emitted particles.

The summing of beta-electron emission will be the biggest challenge, because the energy grid neat the E_max values seems to be unique for every nuclide.

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Thanks!!

Johann van Rooyen

Benchmark CODE 1 against the following hand-calculated SDEF for Cf-252. Here, only MODE N P E is used, because the transport is through 7 cm PB so that FF, AR and A will be radiologically irrelevant. Note that CODE 1 must make no such approximations -- the end-user will delete irrelevant source particle specifiers particles by hand. Reason: Heat deposit calculations or dose rates to lung tissue from an inhaled radionuclide may be requested, and then AR, A and FF become radiologically relevant particles.

```
#       SI7      SP7                DS10
        L        D                  S
        N        0.11649419         21          $ Neutrons
        P        0.56712786         22          $ Photons
        E        0.37439025         23          $ Line-Electrons
        E        0.19223900         24          $ Beta-Electrons (-)
c
c =====
c
c Energy Spectrum of Neutrons Cf-252 SF
SP21    -3    1.18000   1.03419   $ Watt Fission Neutron Spectrum: Cf-252
c
c =====
c
c Line Photons: Cf-252
#       SI22               SP22
        L                  D
        9.9100000E+00      4.2216400E-06
        8.9100000E+00      1.2383500E-05
        7.9100000E+00      3.6587600E-05
        6.9100000E+00      1.0976300E-04
        5.9100000E+00      3.2647400E-04
        5.2500000E+00      3.1666300E-04
        5.2300000E+00      3.5743200E-04
        4.7500000E+00      6.6666000E-04
        4.7300000E+00      6.1354500E-04
        4.2500000E+00      1.3999900E-03
        4.2300000E+00      1.0610400E-03
        3.7500000E+00      2.8666400E-03
        3.7300000E+00      1.8293800E-03
        3.2500000E+00      3.9999600E-03
        3.2300000E+00      3.1521600E-03
        2.8000000E+00      5.3332800E-03
        2.7300000E+00      5.4318500E-03
        2.4000000E+00      9.3332400E-03
        2.2300000E+00      9.4001900E-03
        2.0000000E+00      1.2666500E-02
        1.7300000E+00      1.6183000E-02
        1.5600000E+00      2.3999700E-02
        1.4500000E+00      4.5593700E-03
        1.3500000E+00      5.4599900E-03
        1.2500000E+00      6.5294700E-03
        1.1490000E+00      7.7959600E-03
```

1.1000000E+00	3.9332900E-02
1.0490000E+00	9.3439000E-03
9.4900000E-01	1.1145100E-02
8.4900000E-01	1.3340400E-02
8.4000000E-01	9.3332400E-03
7.4900000E-01	1.5901500E-02
7.4000000E-01	1.1999900E-02
6.4900000E-01	1.9025500E-02
6.4000000E-01	1.5333200E-02
5.7500000E-01	9.8786400E-03
5.4000000E-01	1.9666500E-02
5.2500000E-01	9.8786400E-03
4.7500000E-01	9.8786400E-03
4.4000000E-01	2.5333100E-02
4.2500000E-01	9.8786400E-03
3.7500000E-01	9.8786400E-03
3.7500000E-01	8.9999100E-03
3.2500000E-01	9.8786400E-03
3.2500000E-01	8.9999100E-03
2.7500000E-01	9.8786400E-03
2.7500000E-01	8.9999100E-03
2.2500000E-01	9.8786400E-03
2.2500000E-01	8.9999100E-03
2.0690000E-01	3.0935000E-07
1.9000000E-01	3.9402000E-03
1.9000000E-01	3.6666300E-03
1.7000000E-01	3.9402000E-03
1.7000000E-01	3.6666300E-03
1.5450000E-01	5.0392200E-06
1.5000000E-01	3.6666300E-03
1.5000000E-01	3.9402000E-03
1.3000000E-01	3.6666300E-03
1.3000000E-01	3.9402000E-03
1.2883300E-01	9.8497400E-10
1.2882000E-01	5.4056200E-10
1.2873500E-01	1.3968700E-10
1.2872400E-01	1.2623700E-10
1.2862000E-01	6.3905000E-09
1.2854500E-01	3.1440600E-09
1.2799500E-01	6.4209600E-10
1.2794000E-01	5.7490200E-10
1.2766600E-01	2.5710800E-08
1.2736700E-01	1.2506500E-08
1.2484500E-01	2.0991800E-09
1.2462300E-01	1.8900900E-09
1.2403100E-01	9.7103500E-08
1.2291400E-01	4.9326100E-08
1.1000000E-01	3.6666300E-03
1.1000000E-01	3.9402000E-03
1.0986600E-01	4.2317800E-07
1.0508400E-01	2.7038200E-07
1.0020000E-01	1.2598000E-04
9.5300000E-02	1.1848700E-03

9.0000000E-02	3.6666300E-03
8.5300000E-02	8.3025600E-04
7.5300000E-02	5.8540100E-04
7.0000000E-02	3.6666300E-03
5.5300000E-02	2.8707200E-04
5.0000000E-02	3.6666300E-03
4.5300000E-02	2.0151300E-04
4.3399000E-02	1.4826900E-04
3.1400000E-02	6.9516400E-03
3.0000000E-02	3.6666300E-03
2.4526900E-02	1.5171400E-06
2.4514300E-02	1.8241900E-06
2.4428800E-02	4.2658400E-07
2.4417600E-02	2.6220700E-07
2.4314400E-02	9.7746100E-06
2.4239200E-02	1.0563700E-05
2.3769800E-02	8.6718200E-07
2.3762500E-02	2.6081100E-06
2.3749000E-02	4.5334600E-07
2.3715300E-02	1.0199600E-05
2.3688600E-02	2.1880200E-06
2.3639500E-02	1.1317300E-03
2.3634400E-02	1.3601700E-06
2.3536400E-02	2.9695800E-06
2.3391700E-02	4.9189500E-05
2.3359600E-02	3.7774900E-05
2.3273800E-02	3.8050400E-05
2.3060800E-02	4.1575200E-05
2.2856400E-02	5.4525600E-03
2.2581600E-02	1.1549700E-05
2.2111900E-02	1.8383900E-04
2.0538400E-02	1.1417200E-05
2.0316800E-02	7.6799700E-06
1.9724900E-02	1.1729400E-04
1.9538800E-02	2.3213900E-02
1.8987800E-02	1.8812100E-06
1.8982100E-02	2.6741100E-06
1.8980500E-02	5.0129300E-07
1.8967000E-02	1.9842900E-07
1.8954400E-02	1.8314600E-07
1.8947000E-02	3.7051400E-05
1.8933300E-02	2.1347000E-05
1.8868800E-02	9.0929100E-04
1.8857600E-02	9.8669400E-05
1.8754300E-02	1.2836400E-06
1.8679200E-02	1.0622500E-06
1.8609700E-02	1.0476000E-04
1.8608400E-02	1.5162300E-04
1.8506500E-02	4.0198000E-05
1.8491700E-02	7.3314100E-06
1.8128500E-02	4.5031500E-03
1.8074300E-02	4.8910500E-04
1.7799600E-02	5.1270900E-06

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       1.7500800E-02       4.4054200E-06
       1.7462900E-02       6.6612700E-04
       1.7330000E-02       3.9685100E-04
       1.4978400E-02       2.0876900E-02
       1.4800000E-02       2.6652600E-03
       1.4756800E-02       2.3780600E-03
       1.4165000E-02       1.8492900E-05
       1.3048400E-02       2.0173500E-05
       1.2680900E-02       1.5433700E-03
       5.5600000E-03       2.9671400E-05
       4.7819900E-03       3.5766900E-06
       3.6609000E-03       1.0556600E-02
c
c =====
c
c Line Electrons: Cf-252
#      SI23                SP23
       L                   D
       2.0690000E-01       1.7386300E-08
       2.0207100E-01       4.6036500E-08
       1.8790600E-01       4.6101700E-08
       1.8312400E-01       1.0415500E-07
       1.8234600E-01       1.5278300E-08
       1.5450000E-01       1.0076200E-06
       1.4967100E-01       2.6701200E-06
       1.3550600E-01       3.0890000E-06
       1.3072400E-01       5.9359500E-06
       1.2994600E-01       5.5038300E-07
       1.1962700E-01       1.4029200E-09
       1.0125200E-01       9.9082600E-09
       1.0020000E-01       1.8098200E-04
       9.5371000E-02       4.8169300E-04
       8.2900200E-02       1.7065400E-08
       8.1206000E-02       6.4363500E-04
       7.8040000E-02       4.3140200E-08
       7.6424000E-02       1.0138000E-03
       7.5646000E-02       5.6396600E-05
       4.3399000E-02       1.1442800E-02
       3.8570000E-02       3.0666100E-02
       2.5640000E-02       8.7997600E-07
       2.4405000E-02       4.9522800E-02
       1.9623000E-02       5.8488600E-02
       1.8881200E-02       2.0351700E-03
       1.8845000E-02       2.2282500E-03
       1.5063600E-02       1.6373100E-02
       1.1290000E-02       3.0934400E-02
       3.9817300E-03       1.7693900E-03
       3.2725200E-03       2.8101000E-02
       2.6781300E-03       1.2683400E-01
       1.7958200E-03       1.3603700E-02
c
c =====
c Beta Electron (-) Prob Dens Function: Cf252
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#	SI24	SP24
	A	D
	0.00000	2.601E-11
	0.00010	1.312E-12
	0.00011	9.258E-13
	0.00012	9.186E-13
	0.00013	1.134E-12
	0.00014	1.581E-12
	0.00015	2.347E-12
	0.00016	3.603E-12
	0.00018	8.888E-12
	0.00020	2.232E-11
	0.00022	5.583E-11
	0.00024	1.377E-10
	0.00026	3.331E-10
	0.00028	7.882E-10
	0.00030	1.822E-09
	0.00032	4.114E-09
	0.00036	1.952E-08
	0.00040	8.450E-08
	0.00045	4.669E-07
	0.00050	2.294E-06
	0.00055	1.028E-05
	0.00060	4.324E-05
	0.00065	1.760E-04
	0.00070	7.075E-04
	0.00075	2.713E-03
	0.00080	8.146E-03
	0.00085	1.451E-02
	0.00090	1.766E-02
	0.00100	2.146E-02
	0.00110	2.603E-02
	0.00120	2.999E-02
	0.00130	3.241E-02
	0.00140	3.360E-02
	0.00150	3.419E-02
	0.00160	3.487E-02
	0.00180	3.986E-02
	0.00200	4.117E-02
	0.00220	4.137E-02
	0.00240	4.217E-02
	0.00260	4.301E-02
	0.00280	4.361E-02
	0.00300	4.395E-02
	0.00320	4.413E-02
	0.00360	4.427E-02
	0.00400	4.431E-02
	0.00450	4.436E-02
	0.00500	4.441E-02
	0.00550	4.446E-02
	0.00600	4.452E-02
	0.00650	4.456E-02
	0.00700	4.462E-02

0.00750	4.467E-02
0.00800	4.473E-02
0.00850	4.478E-02
0.00900	4.483E-02
0.01000	4.494E-02
0.01100	4.505E-02
0.01200	4.515E-02
0.01300	4.526E-02
0.01400	4.537E-02
0.01500	4.548E-02
0.01600	4.559E-02
0.01800	4.581E-02
0.02000	4.603E-02
0.02200	4.625E-02
0.02400	4.648E-02
0.02600	4.669E-02
0.02800	4.693E-02
0.03000	4.716E-02
0.03200	4.739E-02
0.03600	4.786E-02
0.04000	4.833E-02
0.04500	4.893E-02
0.05000	4.954E-02
0.05500	5.015E-02
0.06000	5.077E-02
0.06500	5.140E-02
0.07000	5.203E-02
0.07500	5.265E-02
0.08000	5.329E-02
0.08500	5.392E-02
0.09000	5.456E-02
0.10000	5.582E-02
0.11000	5.709E-02
0.12000	5.835E-02
0.13000	5.961E-02
0.14000	6.087E-02
0.15000	6.212E-02
0.16000	6.336E-02
0.18000	6.580E-02
0.20000	6.820E-02
0.22000	7.056E-02
0.24000	7.285E-02
0.26000	7.508E-02
0.28000	7.725E-02
0.30000	7.935E-02
0.32000	8.138E-02
0.36000	8.521E-02
0.40000	8.872E-02
0.45000	9.264E-02
0.50000	9.601E-02
0.55000	9.880E-02
0.60000	1.010E-01
0.65000	1.026E-01

0.70000	1.036E-01
0.75000	1.041E-01
0.80000	1.039E-01
0.85000	1.031E-01
0.90000	1.019E-01
1.00000	9.779E-02
1.10000	9.193E-02
1.20000	8.470E-02
1.30000	7.657E-02
1.40000	6.810E-02
1.50000	5.997E-02
1.60000	5.290E-02
1.80000	4.542E-02
2.00000	4.363E-02
2.20000	4.058E-02
2.40000	3.647E-02
2.60000	3.149E-02
2.80000	2.589E-02
3.00000	1.999E-02
3.20000	1.414E-02
3.60000	4.252E-03
4.00000	0.000E+00

c